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COASTAL BLUE CARBON TRAINING

EELGRASS SEDIMENT CORING METHODS

Field methods, lab procedures, and data analysis



Agenda

Day 1 –

Morning = Meeting room

- Background on blue carbon
- Eelgrass coring methods
- Project planning
- Data processing and interpretation

Lunch

Afternoon = **Field site #1**

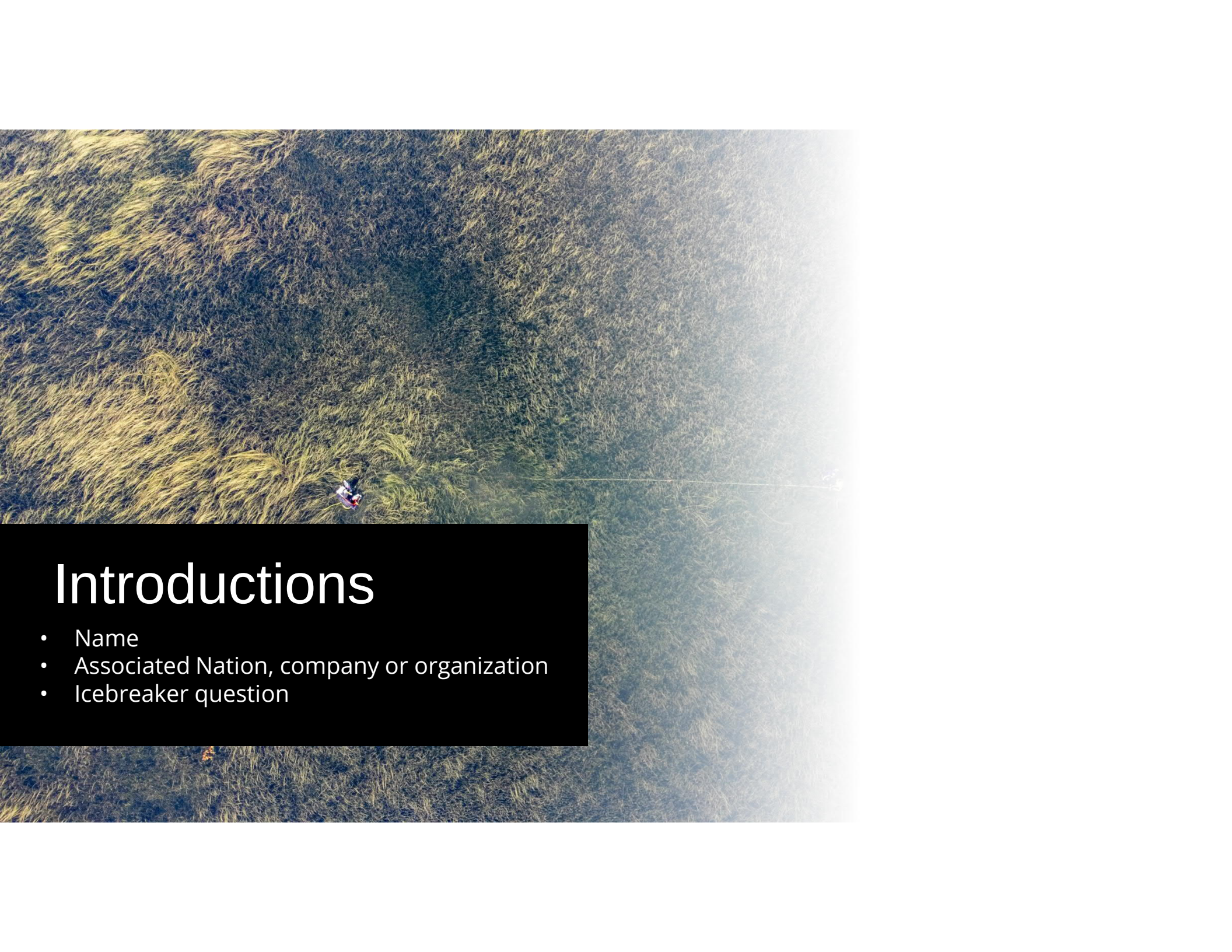
- Hands-on practice with equipment
- Overview of data recording
- Collecting and packaging samples



Agenda

Day 2 - Field site # 2

- Implement an eelgrass coring project in an area of interest
- Core collection, core extraction, laboratory prep
- Sample storage
- Data processing and interpretation



Introductions

- Name
- Associated Nation, company or organization
- Icebreaker question

DAY 1 MORNING

- 1) Brief overview of blue carbon
- 2) Four steps to completing a carbon stock assessment
 - i. Planning
 - ii. Collecting samples
 - iii. Analyzing sampling
 - iv. Interpreting the results
- 3) Applying a project
- 4) After the field work



1) Intro to “Blue Carbon”



'Blue Carbon' Ecosystems

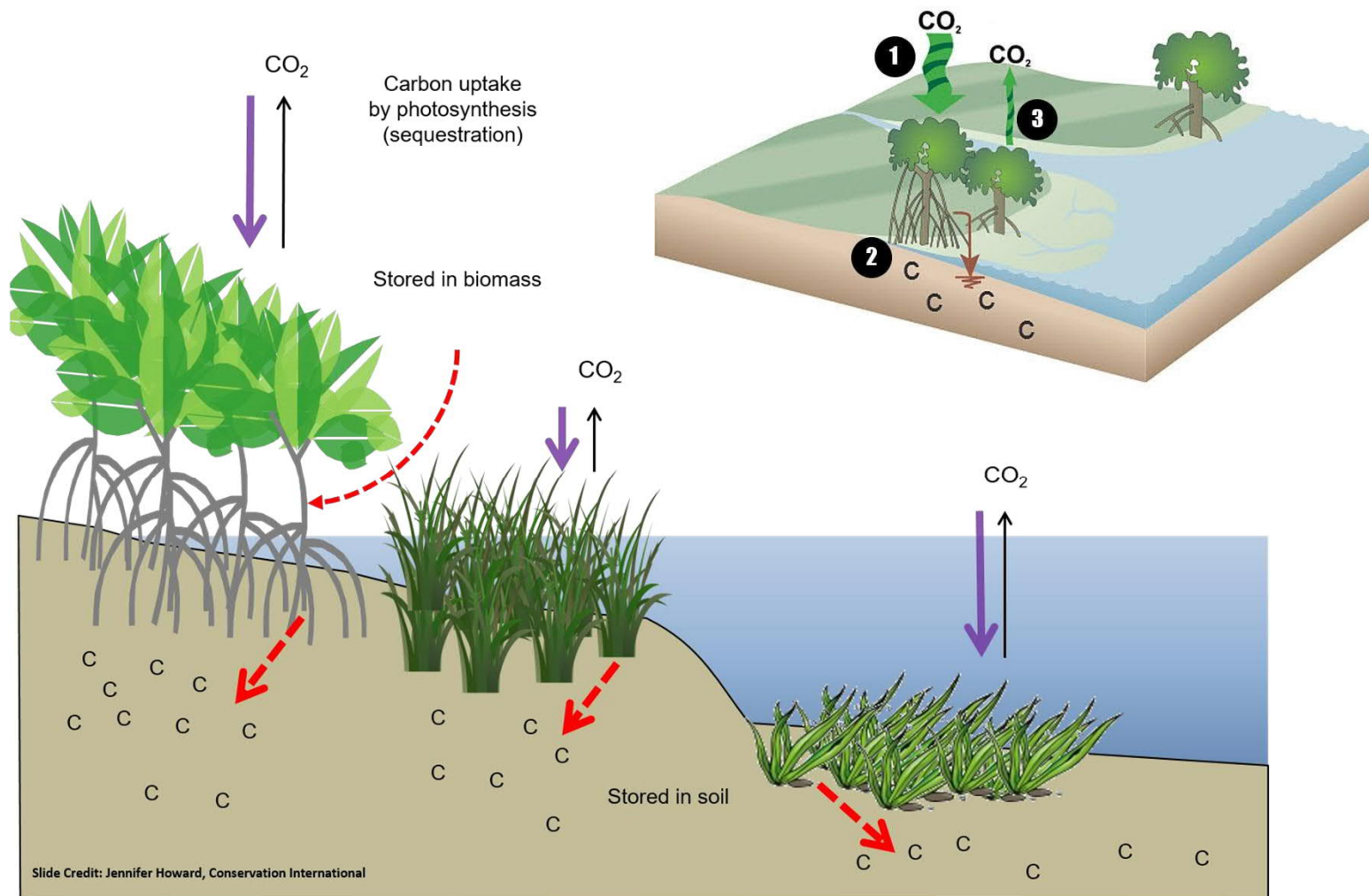
Coastal ecosystems that capture and store carbon in biomass and sediments:

- Mangroves
- Tidal marshes
- Eelgrass/Seagrass meadows

These systems store carbon intensively and long-term, especially below ground.



Blue carbon habitats act as carbon sinks

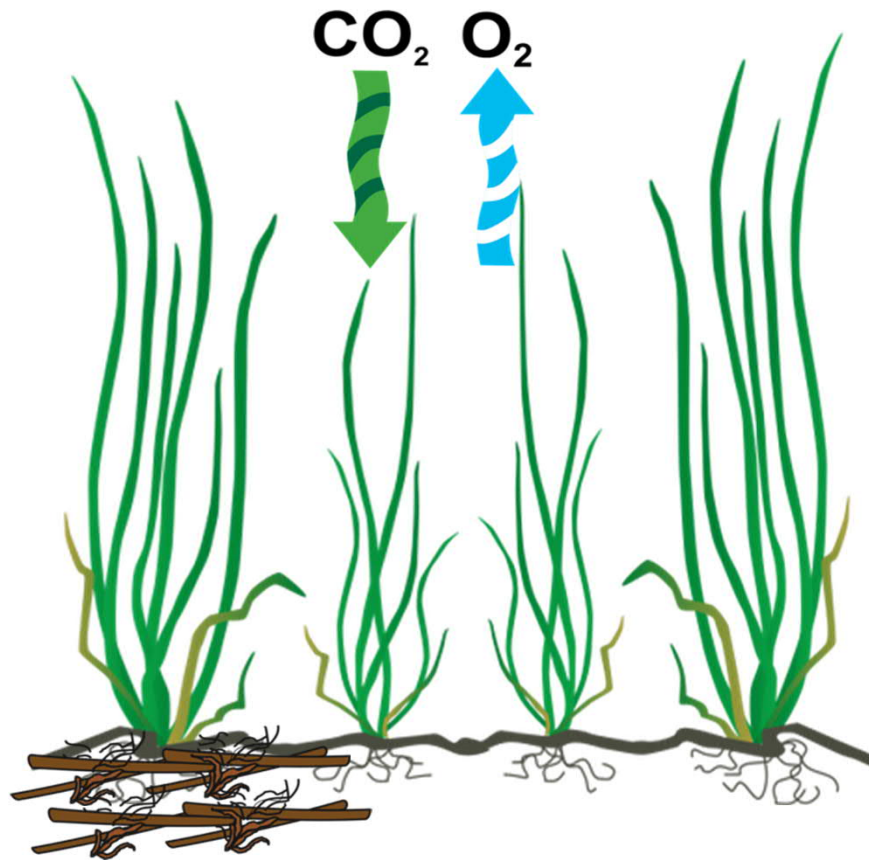


What is carbon cycling?

Plants are taking in CO₂
(from the water)

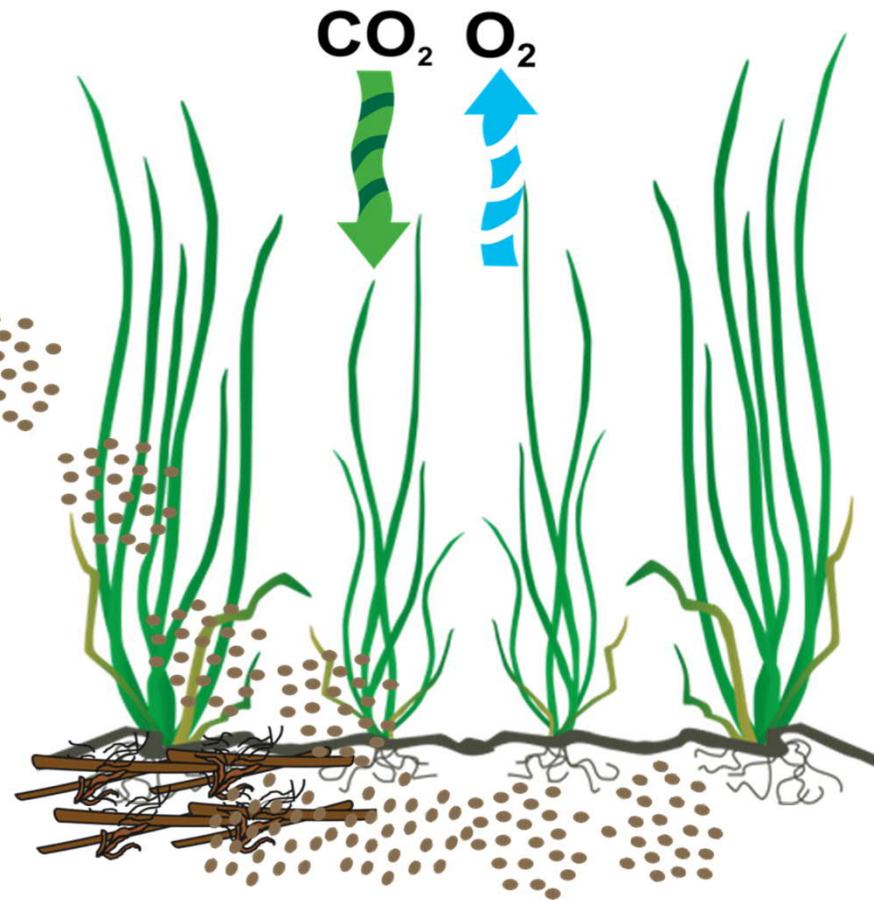
And breathing out O₂

Carbon entering ecosystem
via photosynthesis



What is the carbon cycling?

Weathering and transport of suspended particles into the ecosystem



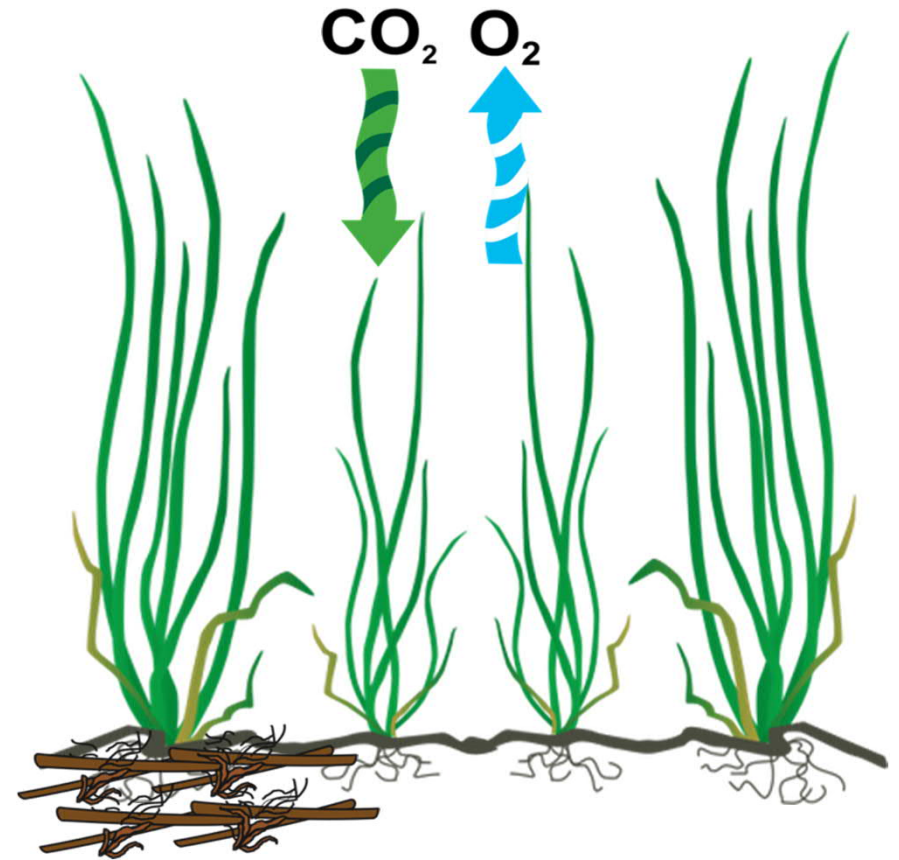
Brought in from tides or upstream from the watershed

Where does the carbon come from?

Autochthonous (Produced in-place)
Seagrass photosynthesis
Root and rhizome inputs

Allochthonous (Transported in)
From tides, rivers, adjacent systems
Trapped in sediments

Seagrass meadows are both carbon
producers and carbon traps.



Where does the carbon come from?

Autochthonous (Produced in-place)

Seagrass photosynthesis

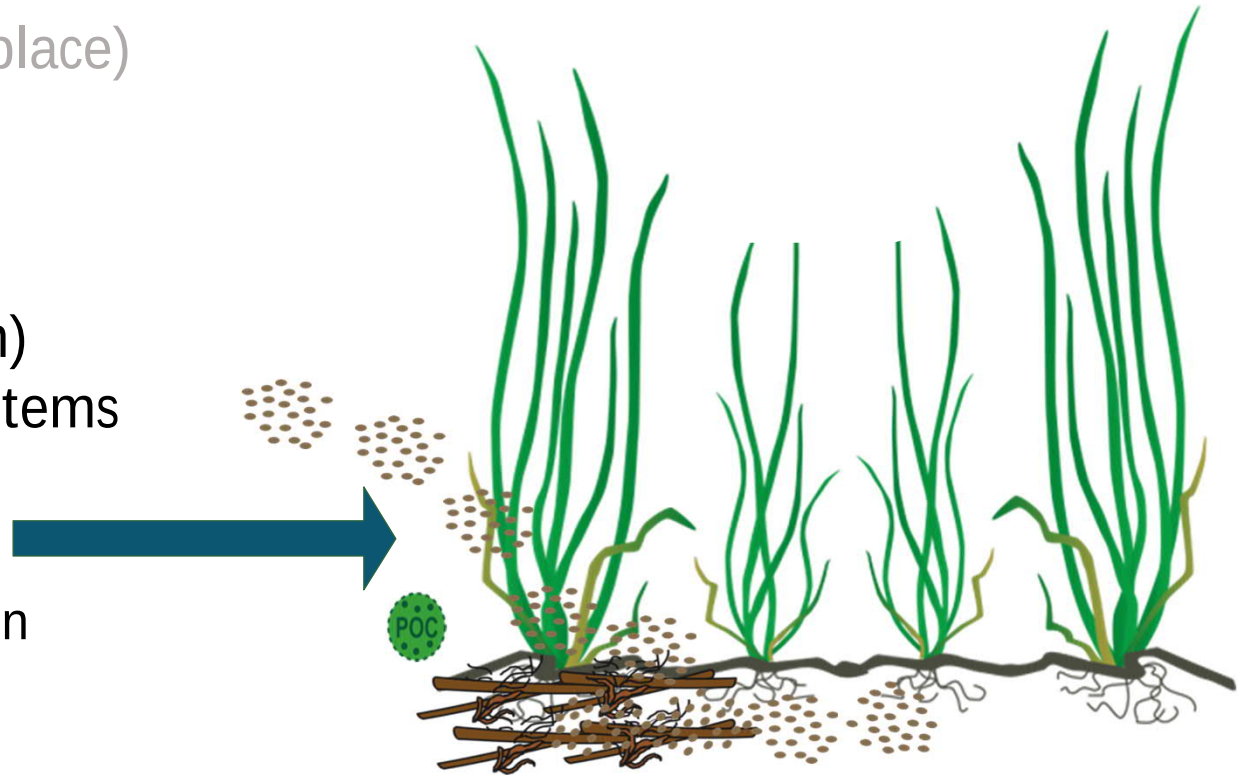
Root and rhizome inputs

Allochthonous (Transported in)

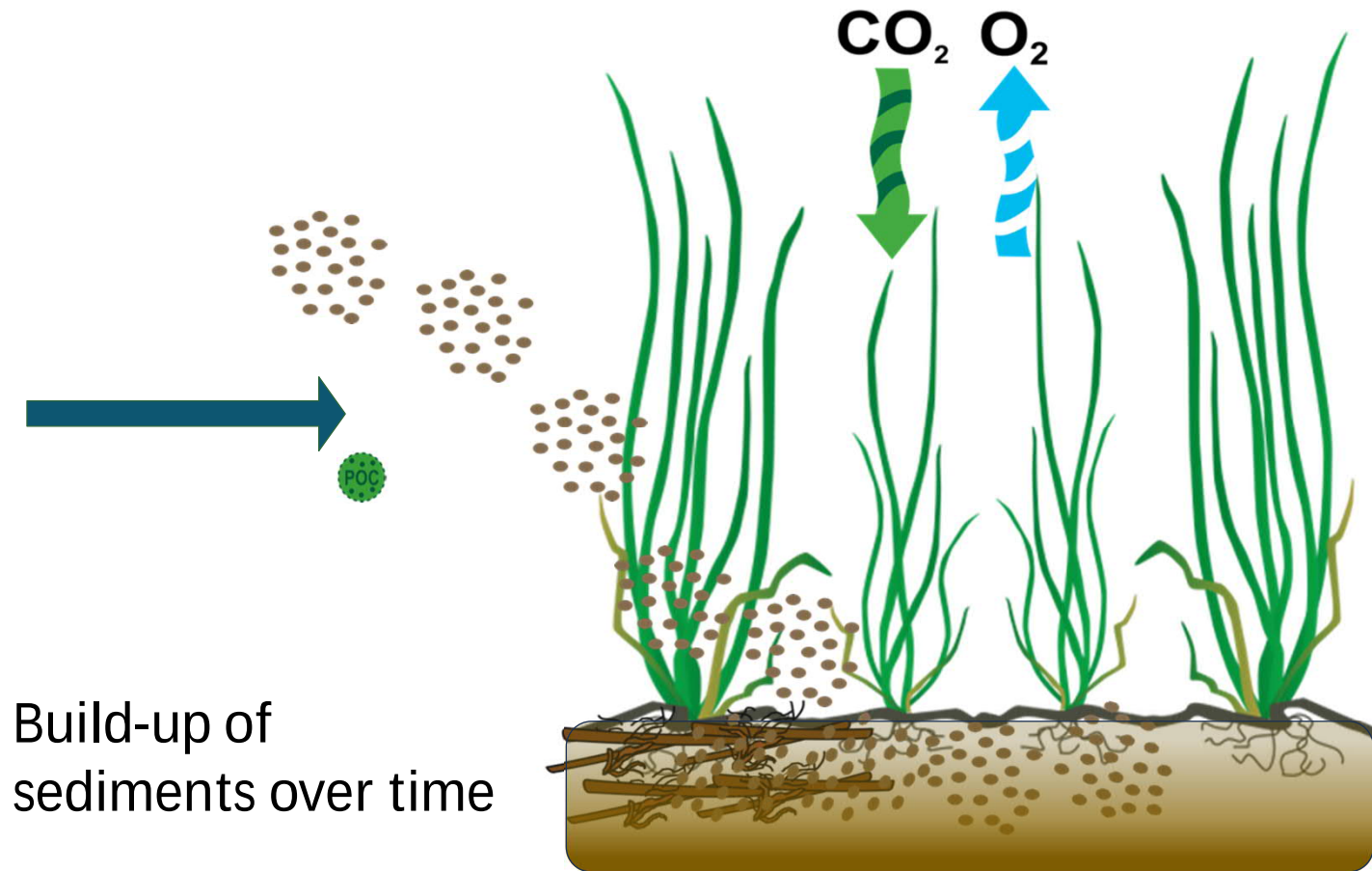
From tides, rivers, adjacent systems

Trapped in sediments

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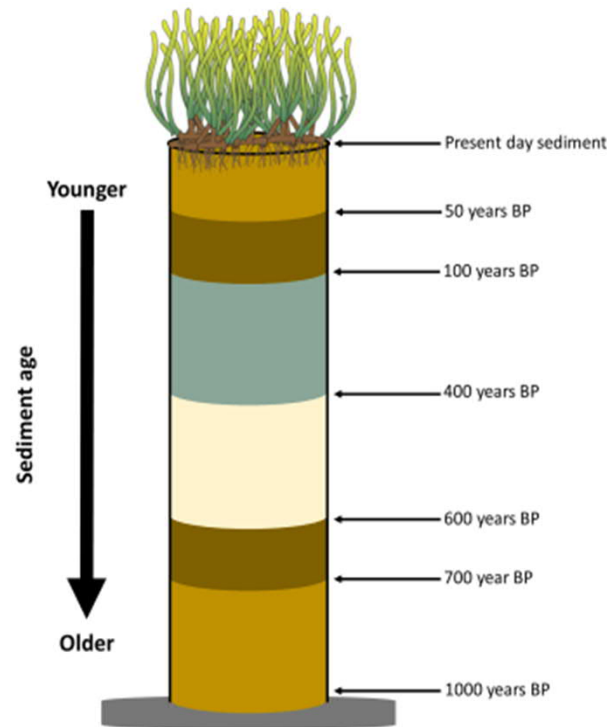
What is the soil carbon cycling?



What is the soil carbon cycling?

What is sediment?

- “Layers of **transported** and **deposited particles** or aggregates derived from rocks, soil, or biological material” (SSSA 2021)



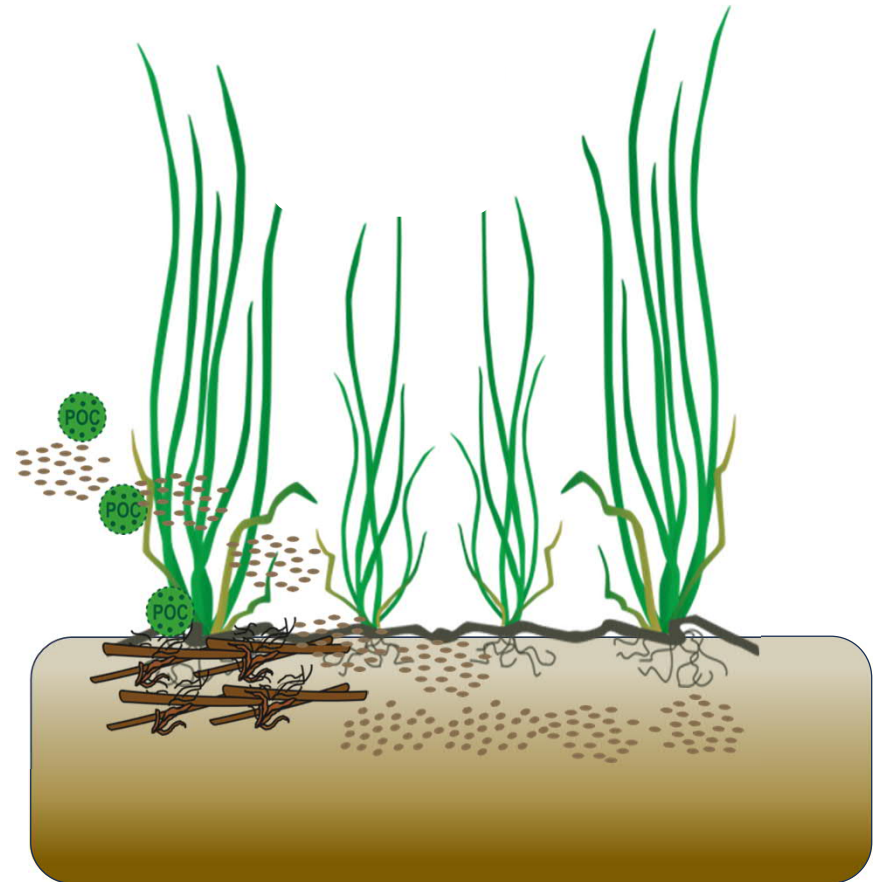
Carbon stocks vs carbon rates

Carbon Stock Total carbon stored at one point in time

Sequestration Rate Carbon stored per year

Seagrass systems = high input and output. Net storage happens in sediments.

Sediment cores measure stocks — but they reflect long-term system dynamics.



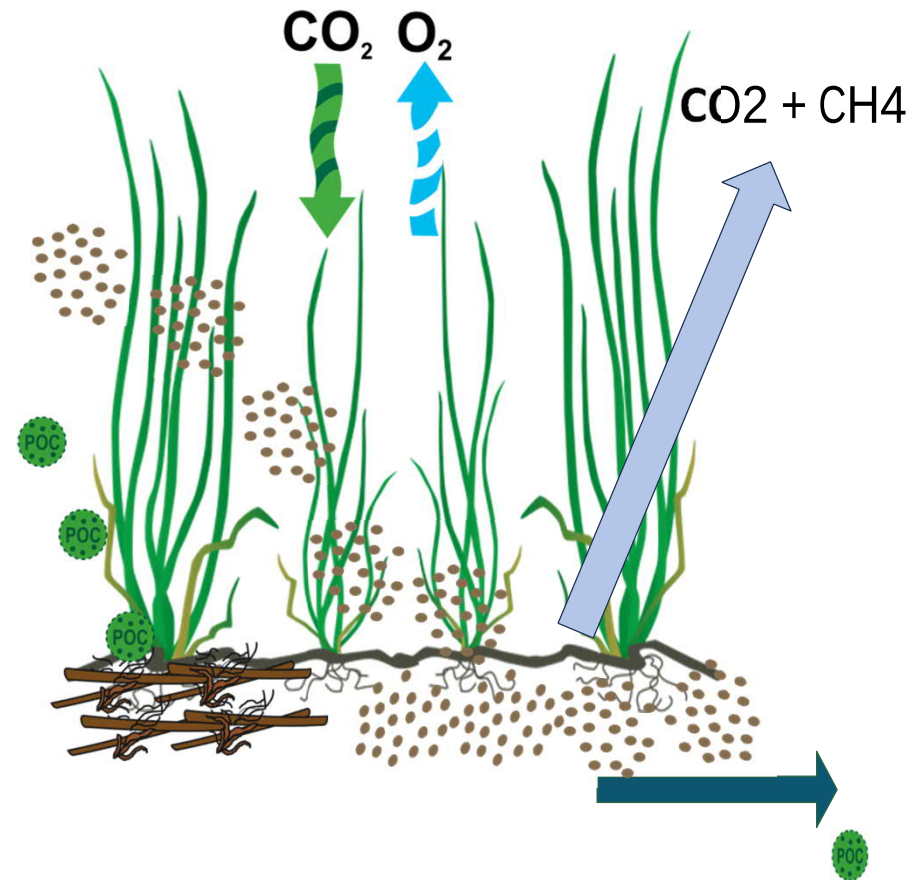
Carbon stocks vs carbon rates

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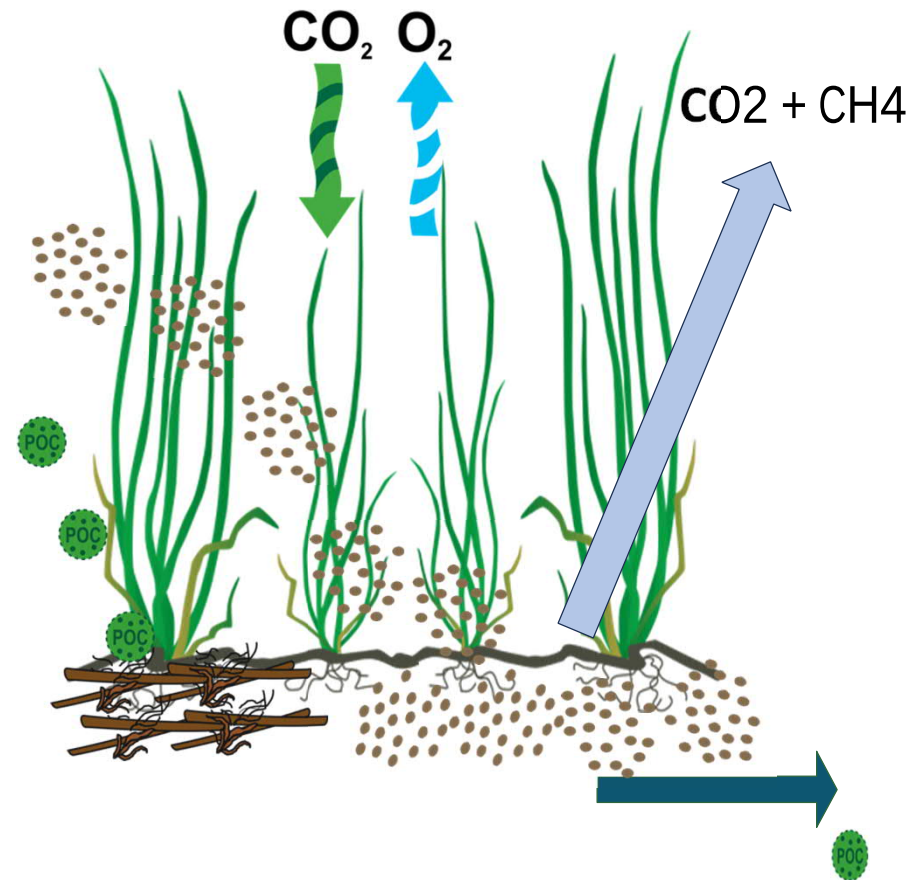
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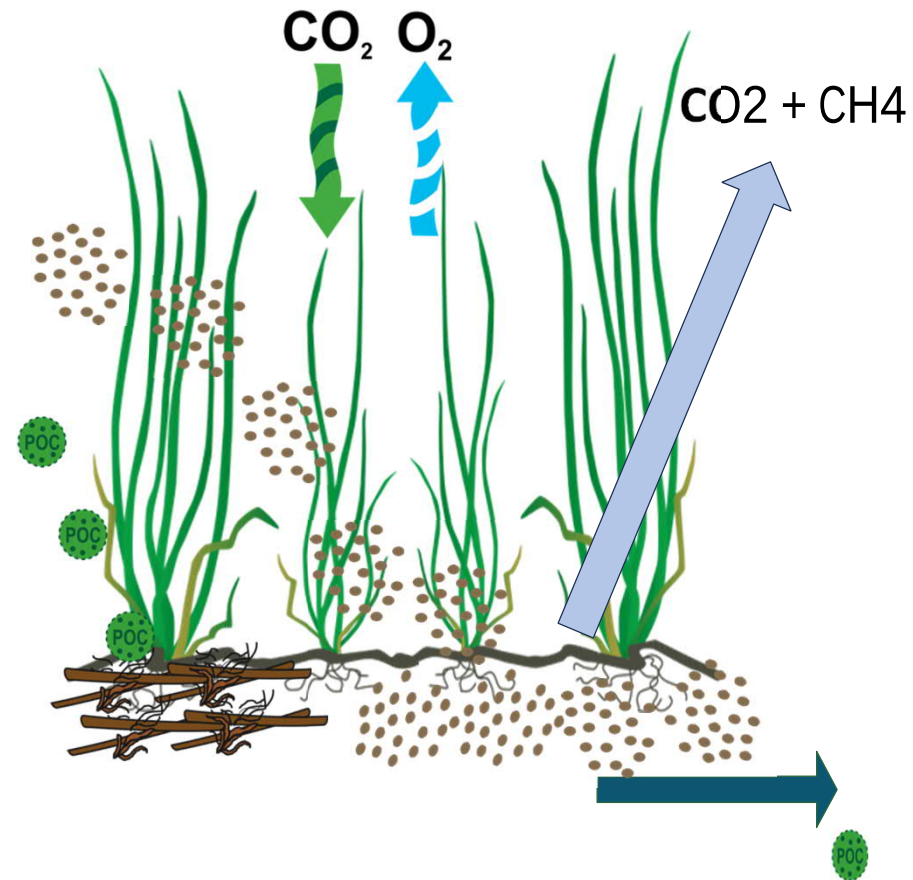
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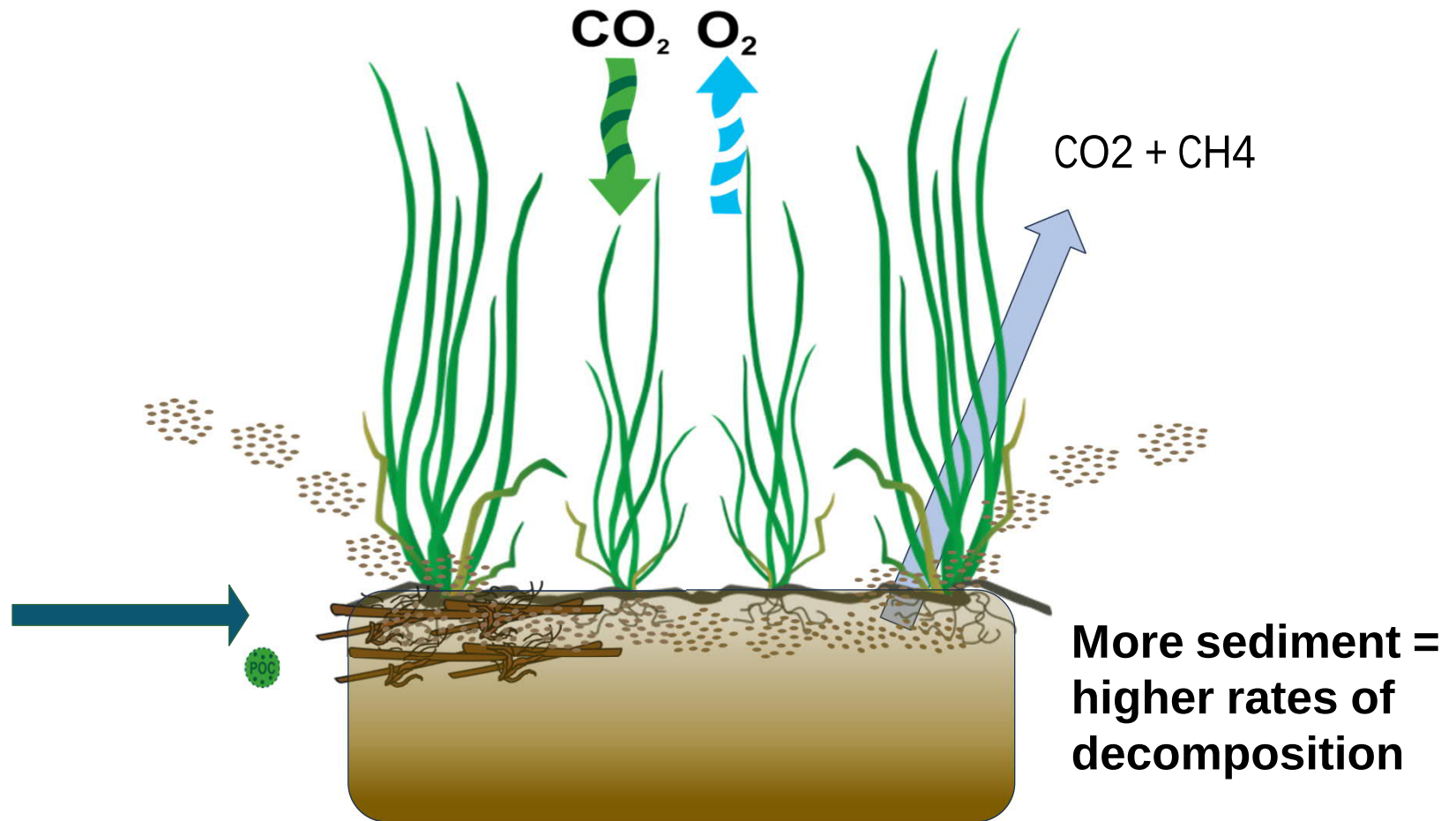
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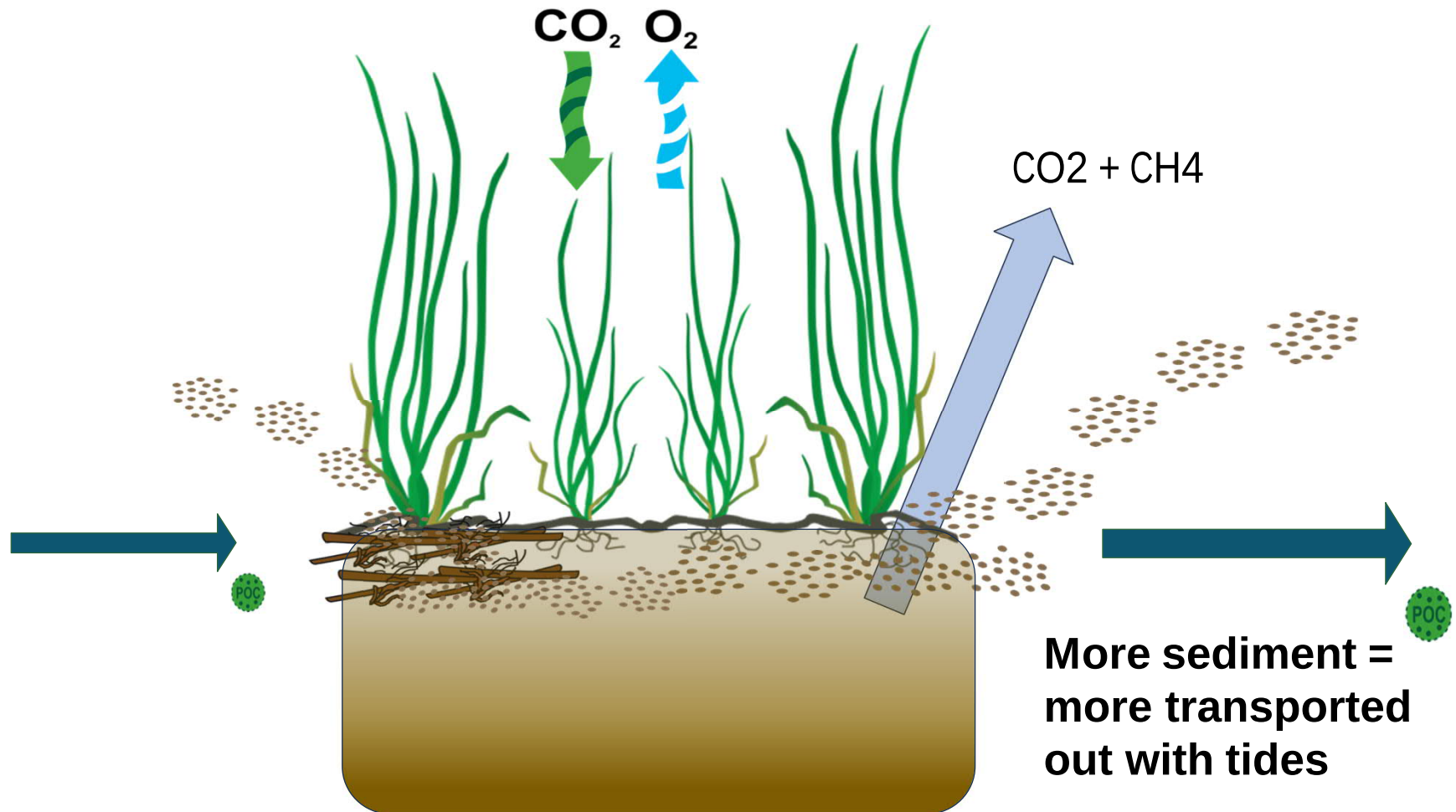
Sediment cores measure stocks which reflect long-term fluxes



What is the soil carbon cycling?



What is the soil carbon cycling?

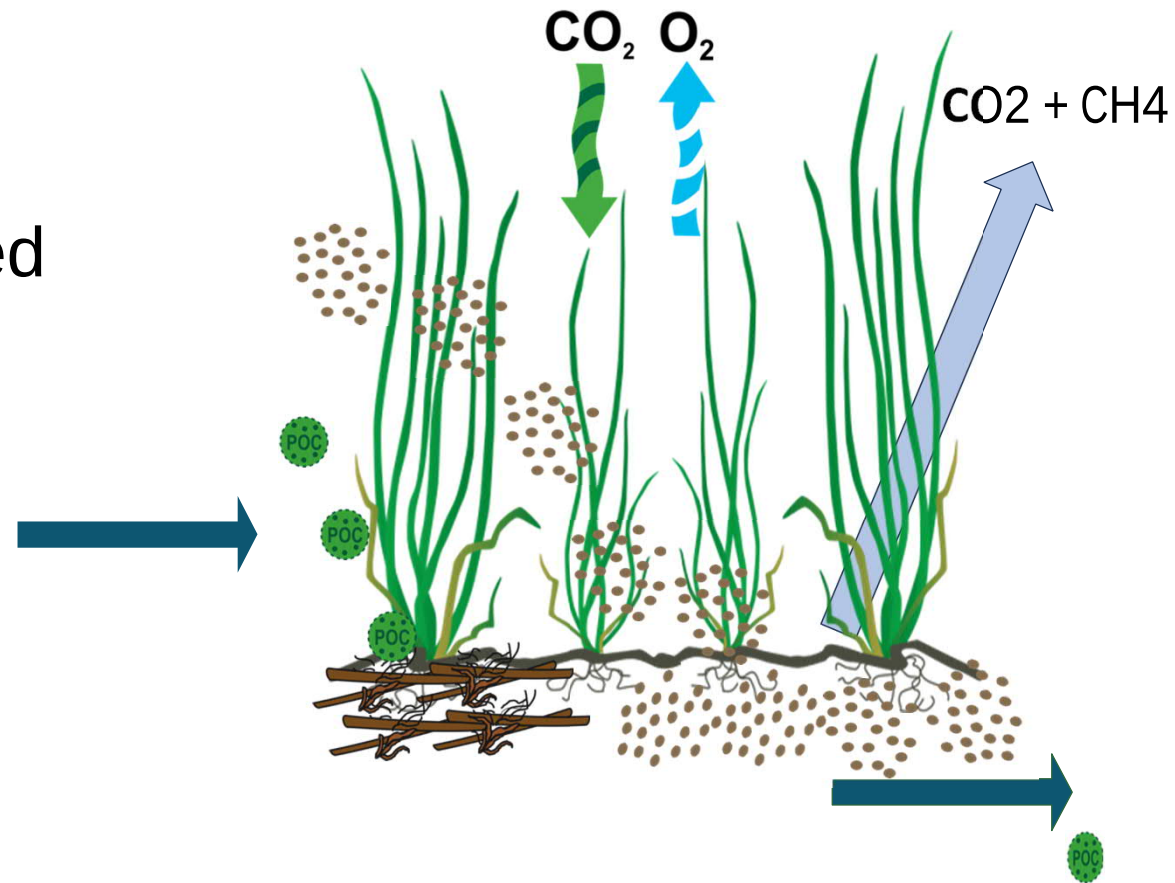


What is the soil carbon cycling?



Net balance

“Is more carbon being added to the soil than is being released?”



Carbon balance in blue carbon systems



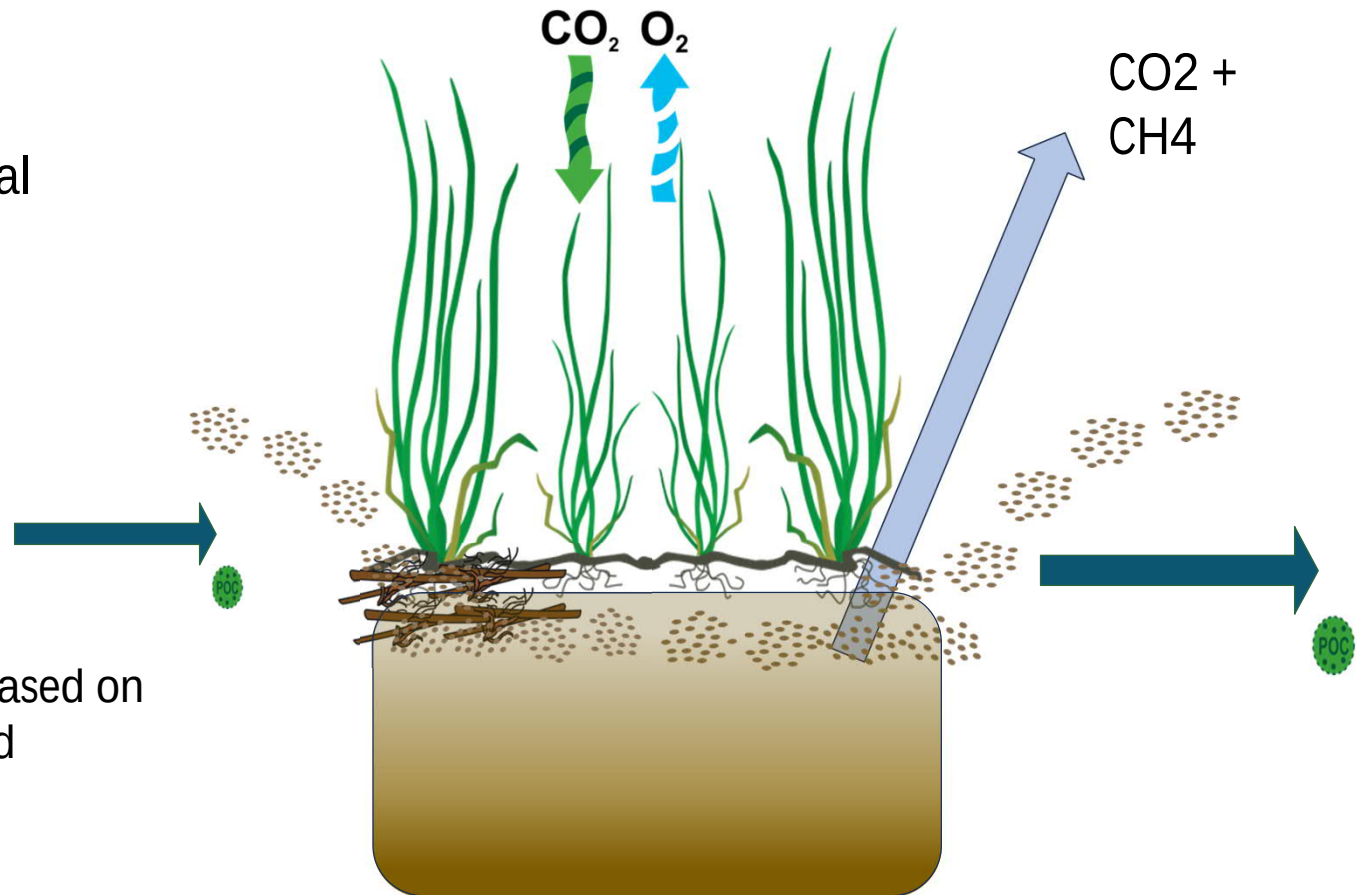
Carbon IN

Photosynthesis
Transported organic material
Sediment inputs

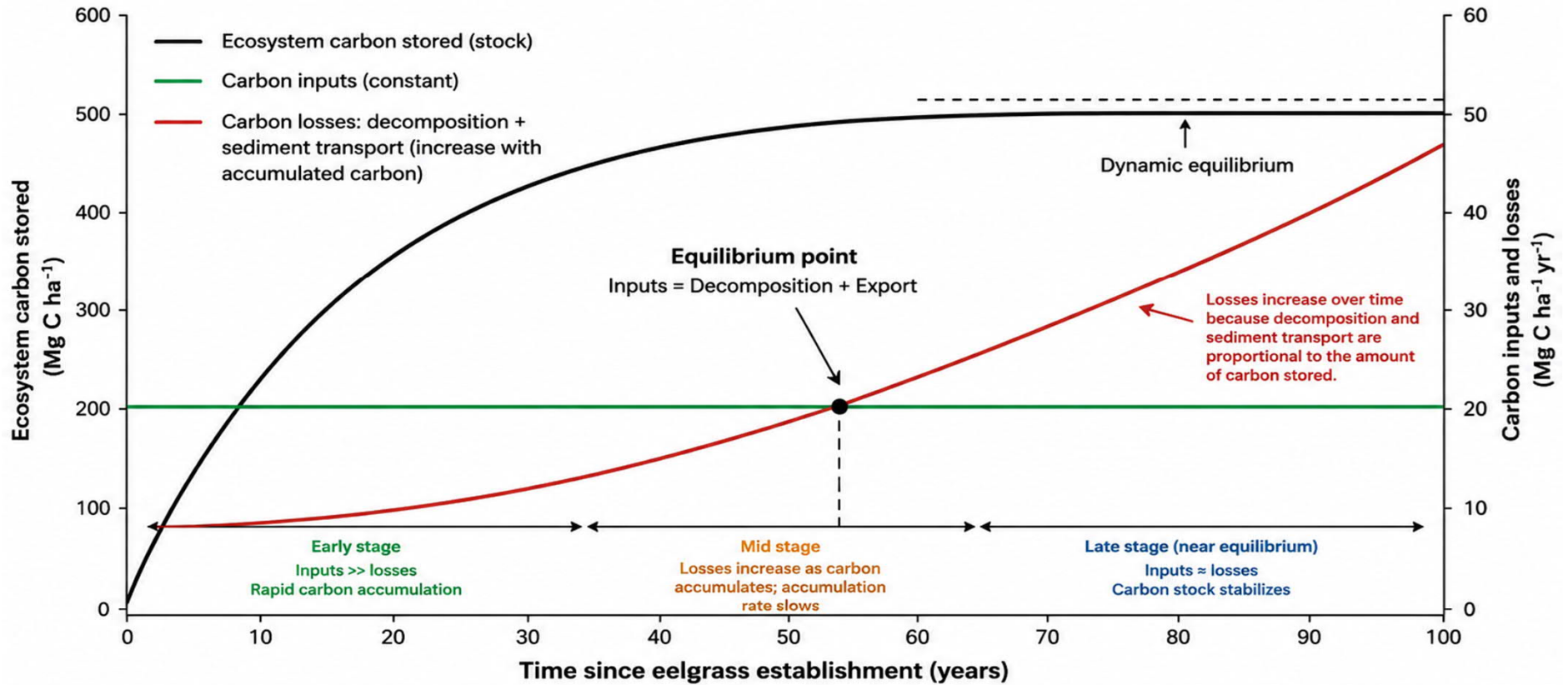
Carbon OUT

Respiration → CO_2
Methane (CH_4)
Export via tides

Blue carbon systems function based on the balance between inputs and outputs.



Carbon accumulation in an eelgrass ecosystem over time



Carbon inputs (constant through time)

- Carbon fixation through photosynthesis
- Allochthonous carbon deposition from water column



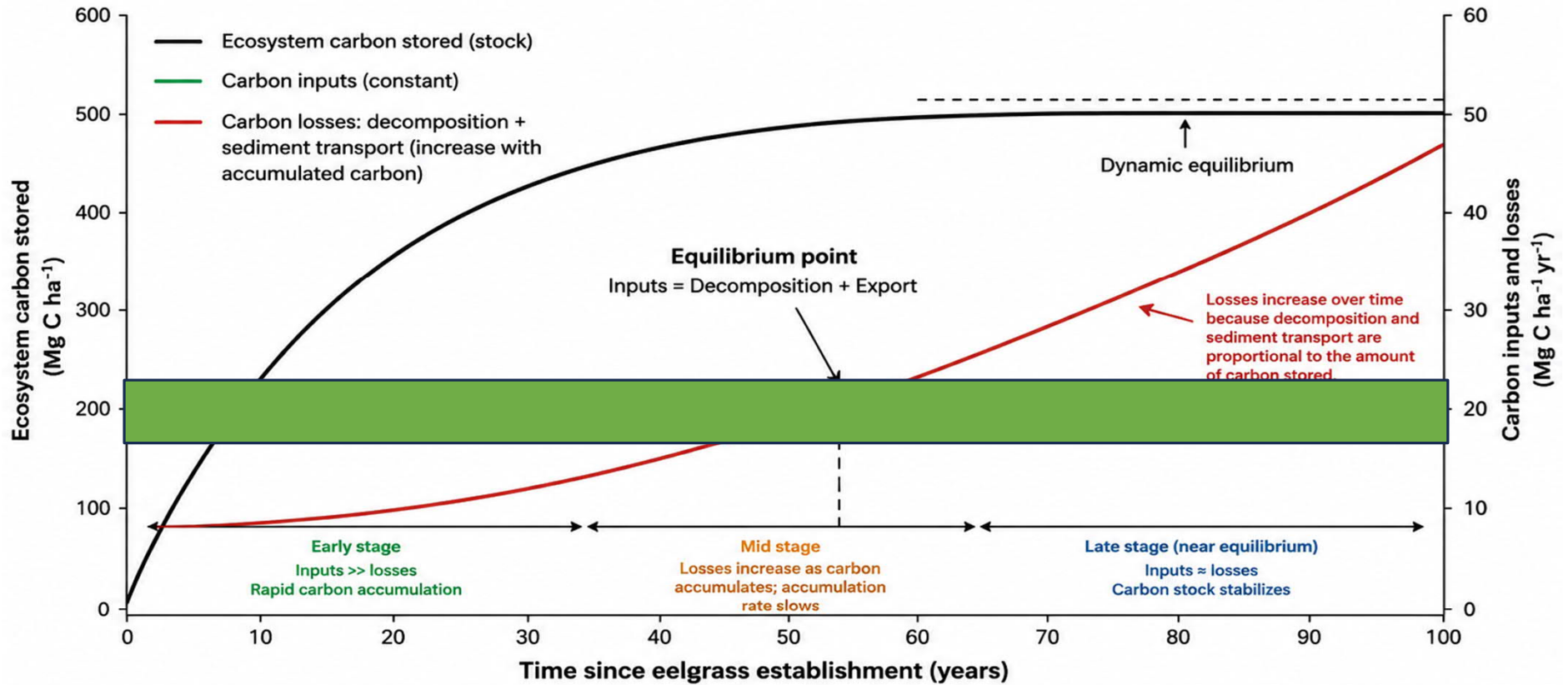
Carbon losses (increase with carbon stored)

- Decomposition of organic matter
- Sediment transport / export (resuspension, burial below active layer, or advection)



Conceptual model: As more carbon is stored in sediments, decomposition and sediment transport increase proportionally, reducing net accumulation until a dynamic equilibrium is reached.

Carbon accumulation in an eelgrass ecosystem over time



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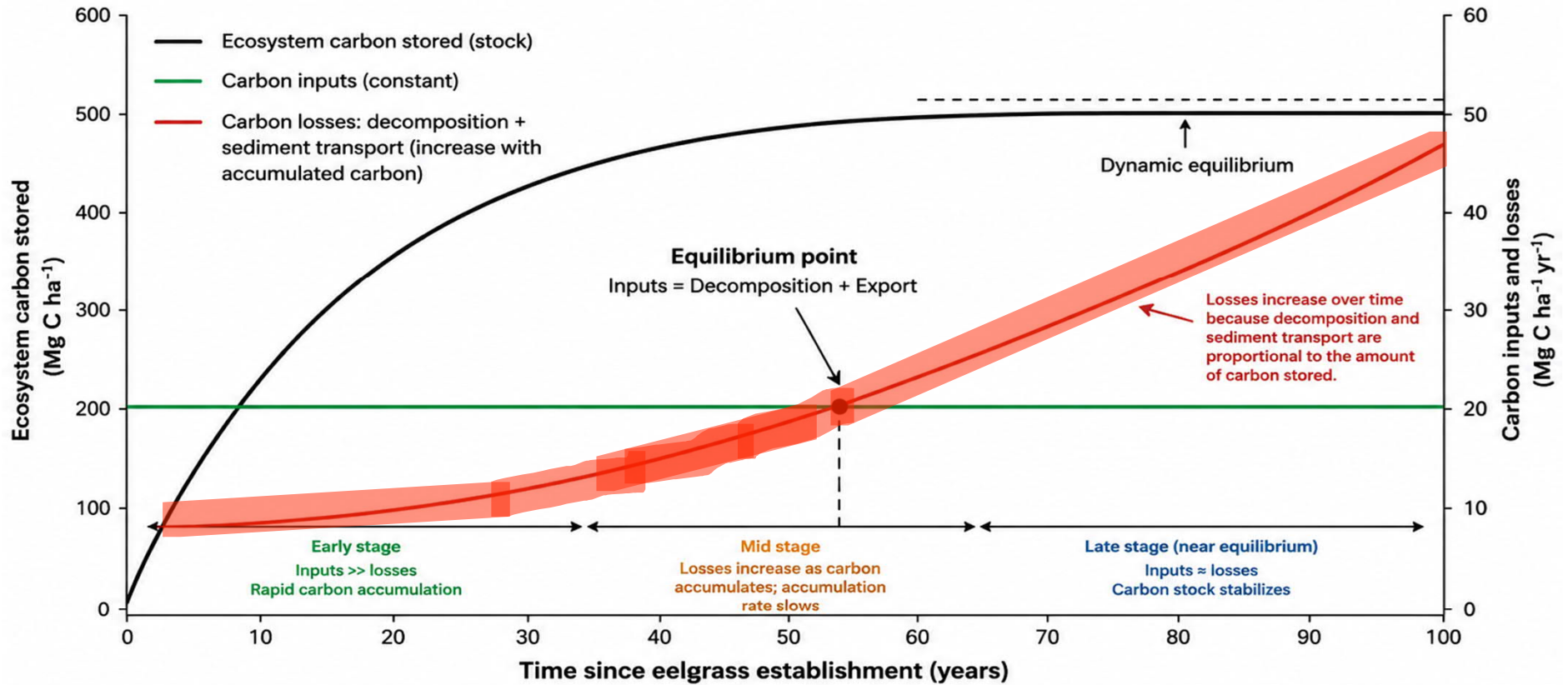
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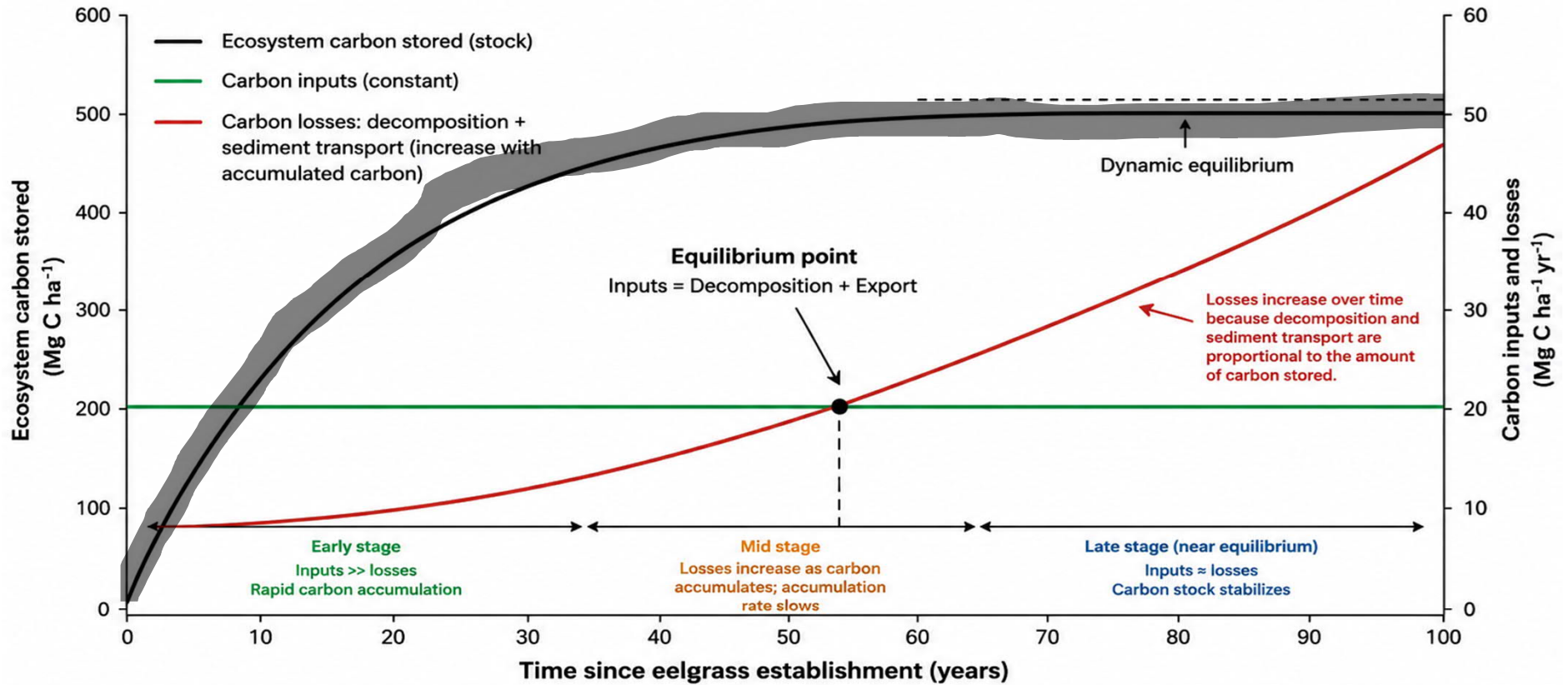
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When Carbon Systems Are Stable vs Disturbed



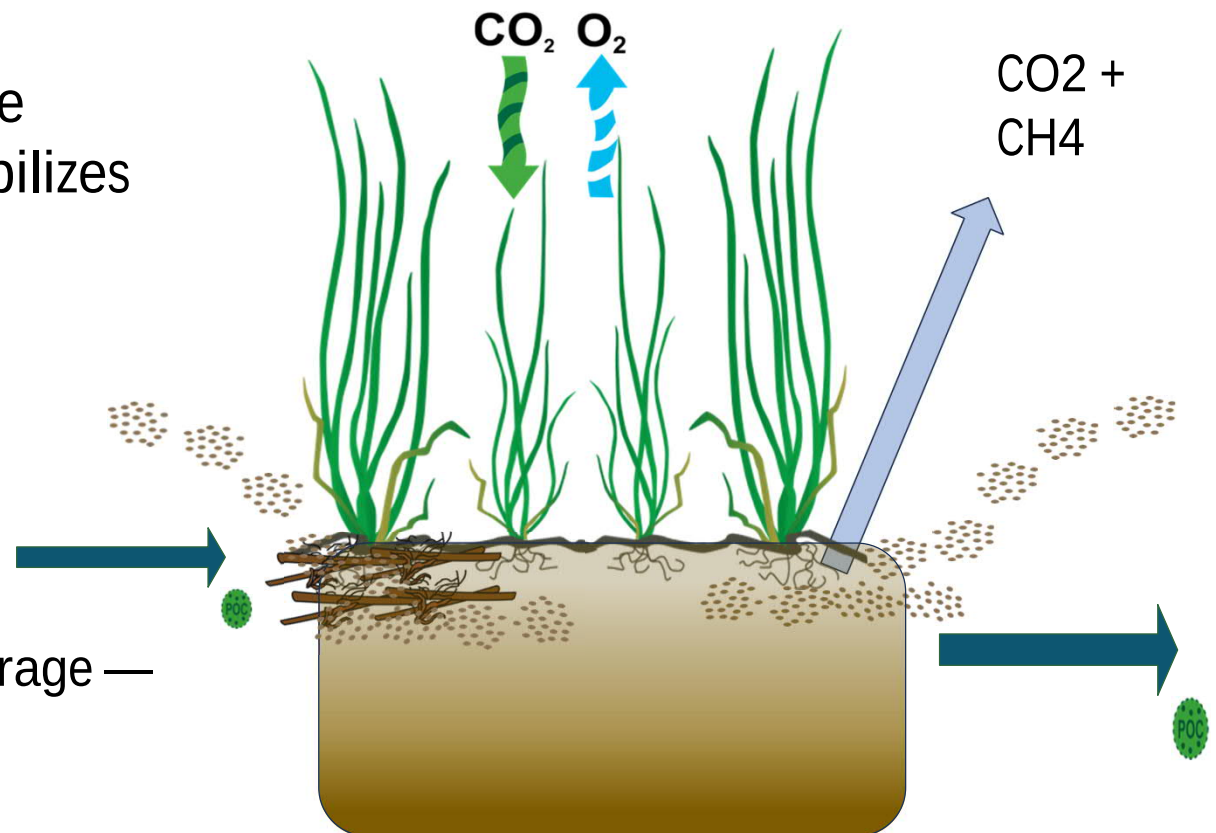
Healthy systems

Carbon accumulates over time
System balance gradually stabilizes

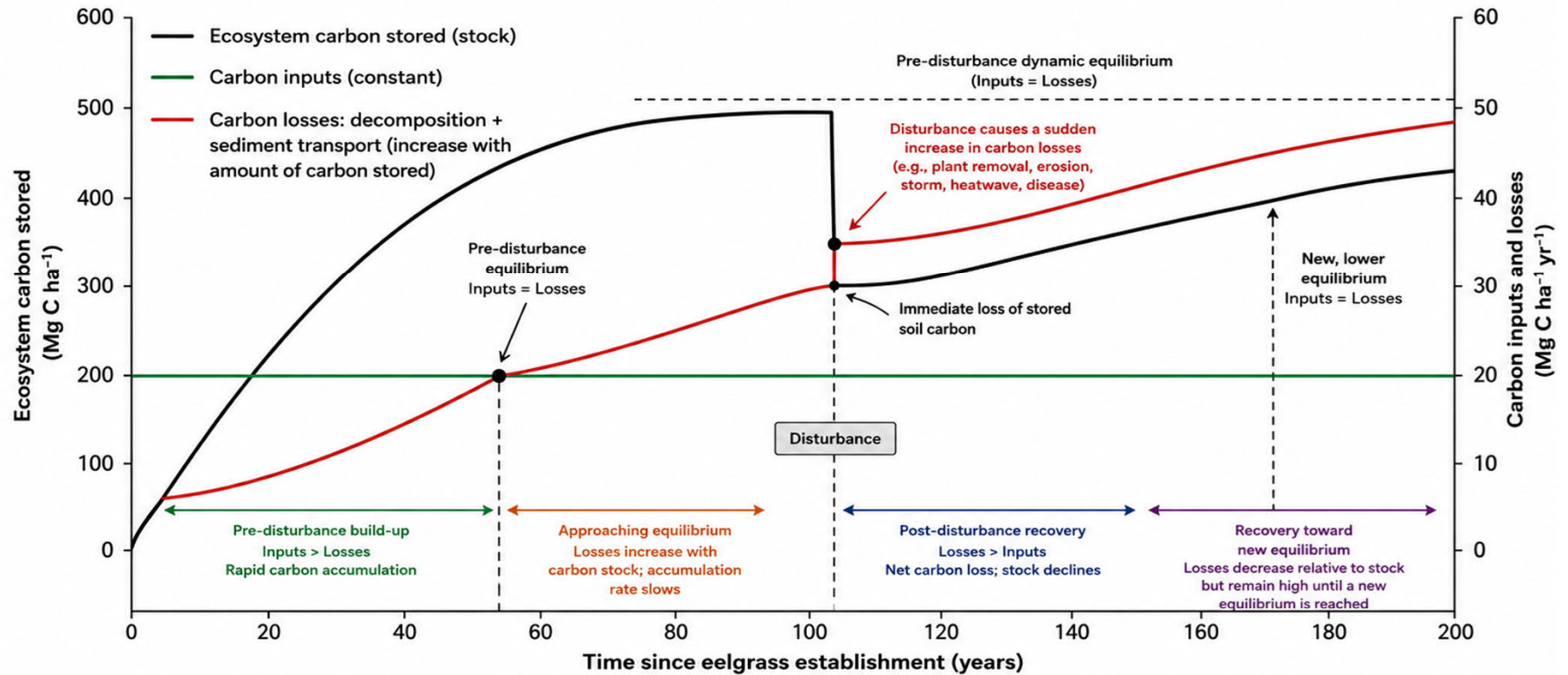
Disturbance

Releases stored carbon
Can shift system from sink → source

Disturbance doesn't just stop storage —
it can reverse it.

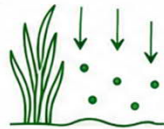


Carbon accumulation in an eelgrass ecosystem over time with a disturbance



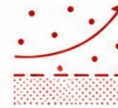
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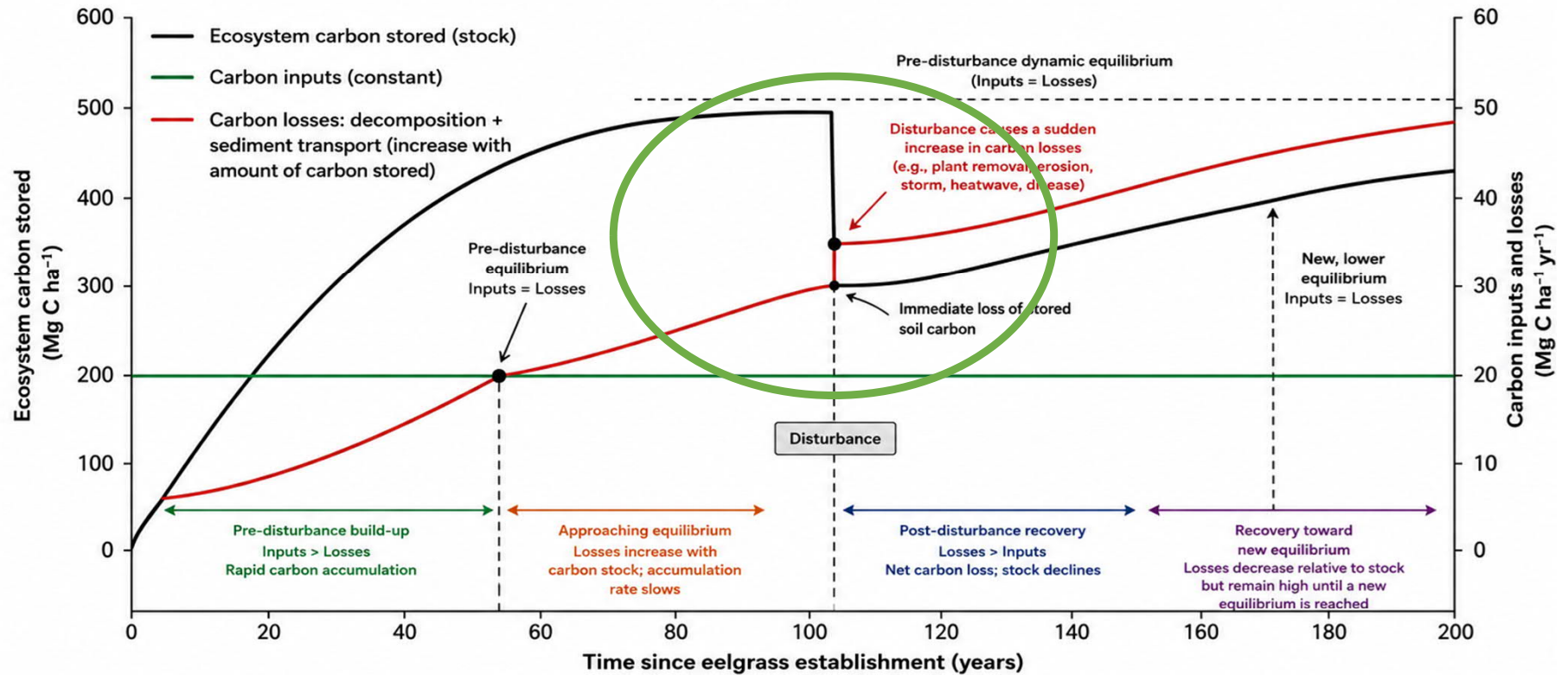
Disturbance examples

- Storm / wave energy
- Physical disturbance (e.g., anchoring, dredging, trawling)
- Heatwave, disease, hypoxia
- Extreme erosion / sediment loss



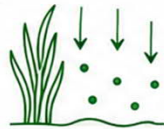
Conceptual model: Disturbance abruptly increases carbon losses and reduces stored carbon. Recovery is slow, with losses remaining elevated until a new, lower equilibrium with inputs is reached.

Carbon accumulation in an eelgrass ecosystem over time with a disturbance



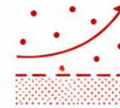
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Carbon losses (increase with carbon stored)

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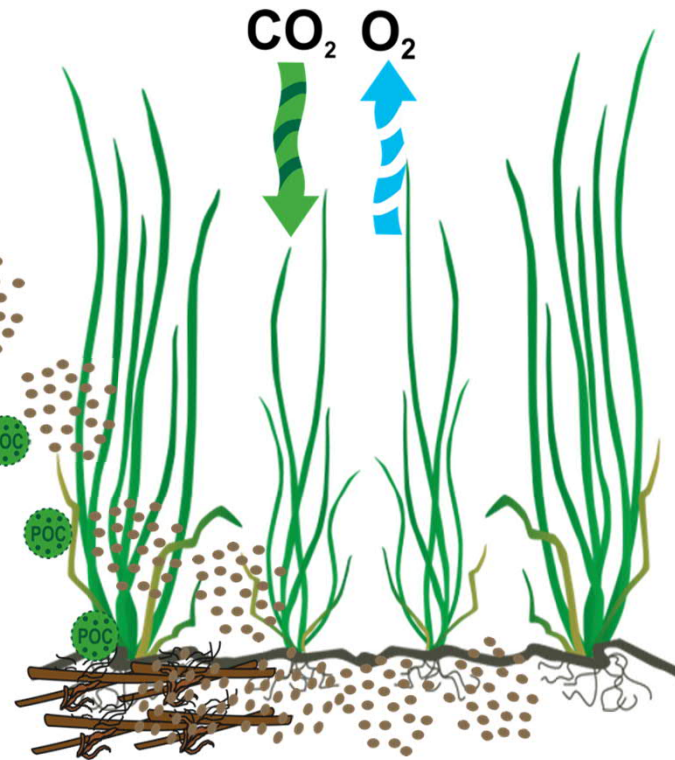
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Recipe for an effective blue carbon ecosystem

High rates of photosynthesis

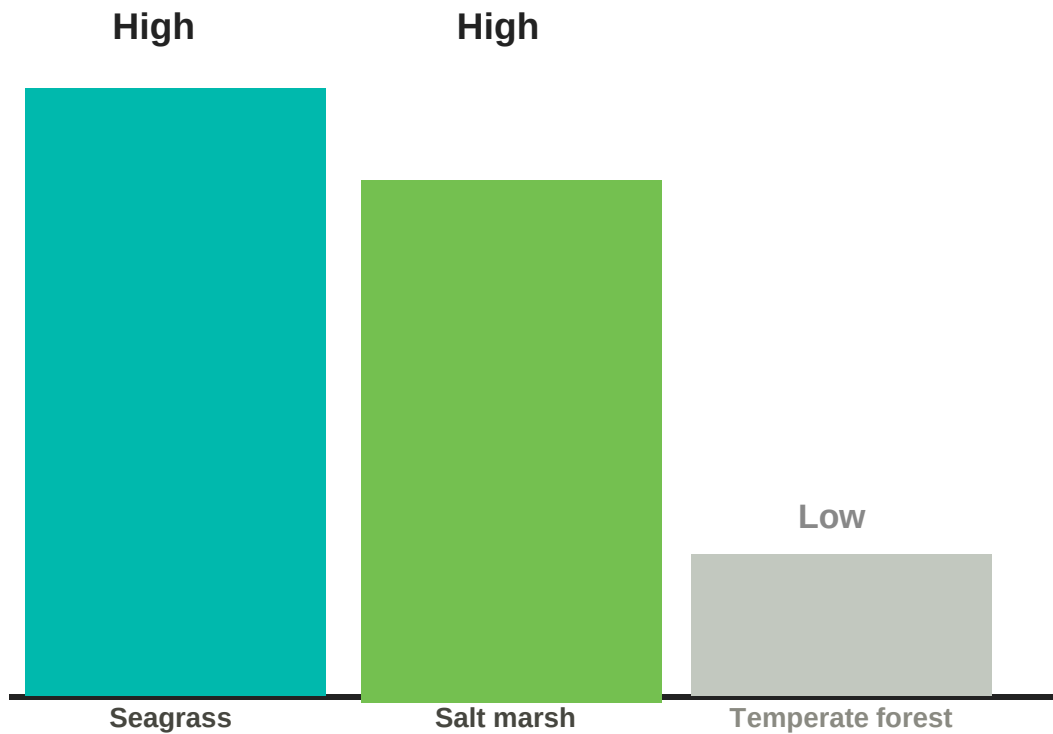
Build up sediments through time

Capture carbon
from surrounding
ecosystems



Low rates of decomposition

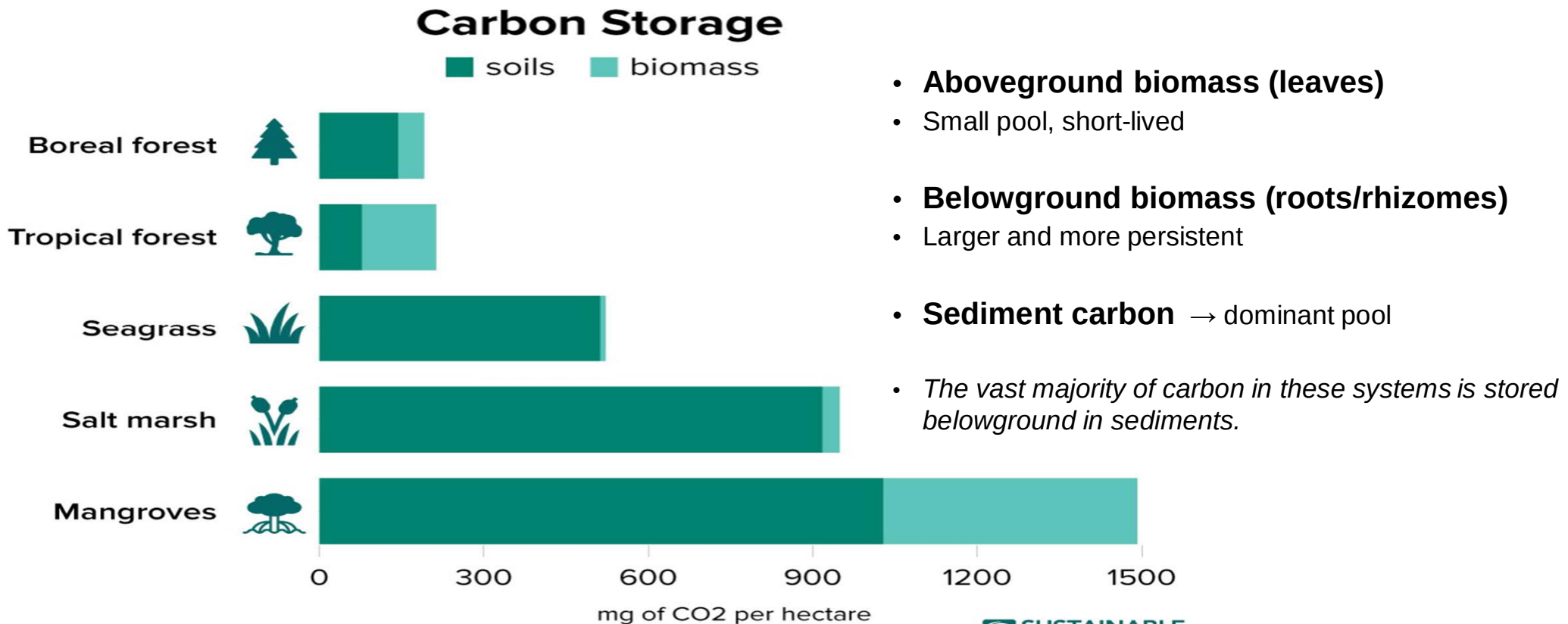
Sediment carbon flux per year



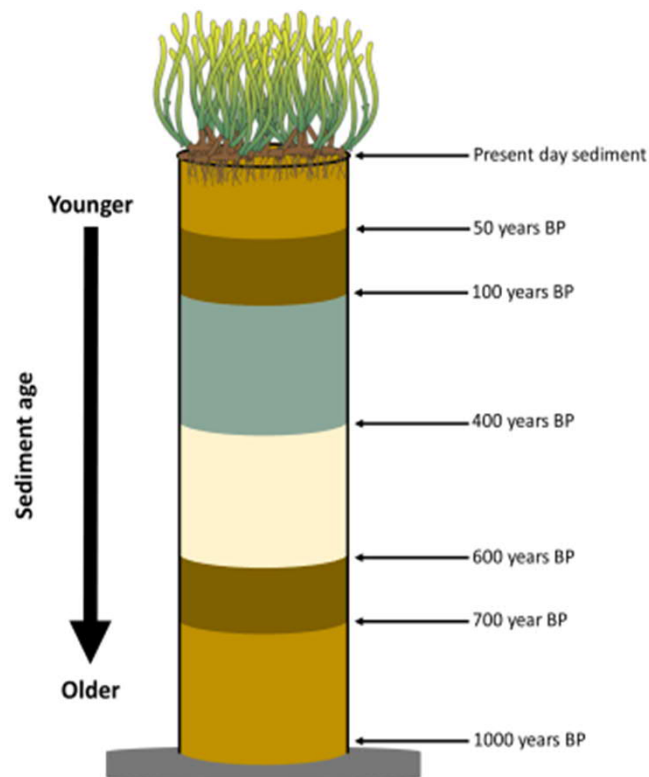
Carbon flux in
Eelgrass/Seagrass is high

“Lot of carbon enters and
exits the ecosystem”

Carbon Pools in Seagrass Ecosystems



(McLeod et al 2011)



The carbon is in the soil

Far more carbon sits below ground in sediment than in the living leaves and roots above it.

It stays locked away

Waterlogged, low-oxygen sediments slow decay, storing carbon for centuries to millennia.

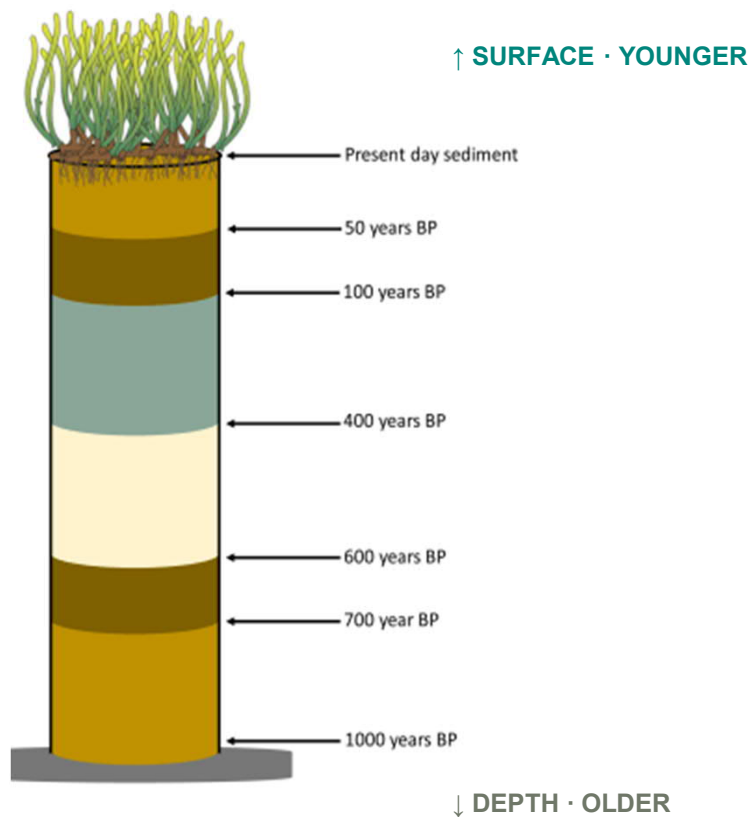
Eelgrass carbon can be lost

When eelgrass is disturbed or destroyed, that stored carbon can be released again.

Carbon accumulates over long time periods — decades to centuries.

Disturbance can release decades–centuries of stored carbon rapidly.

LAYERED TIME CAPSULES

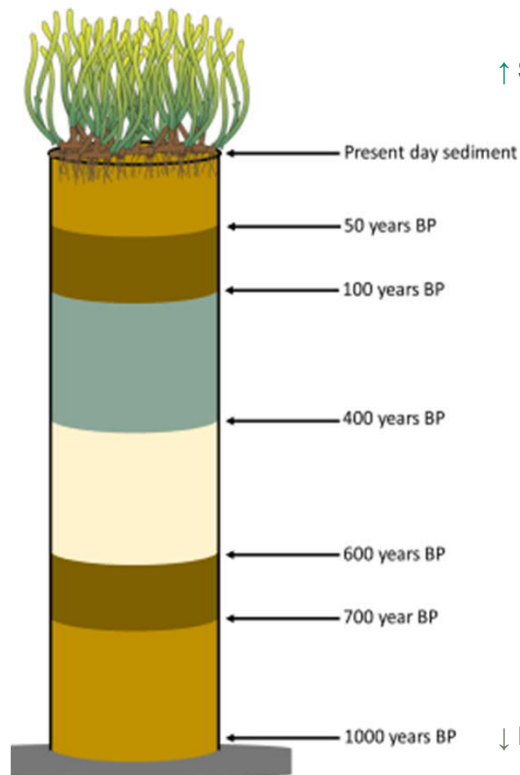


Pull a core out of the ground and you pull out a stack of time. The layers stay in order — **younger on top, older at the bottom.**

Each layer records...

- **Carbon** — how much is stored at that depth
- **Accretion** — how fast sediment built up over time
- **History** — past disturbances, storms, and change

LAYERED TIME CAPSULES



↑ SURFACE · YOUNGER

Typically – higher % of carbon in top of soil, but soils are less dense

- Recent plant
- Roots
- Less time for decomposition

Soils typically have lower % Carbon, but soils are more dense

- More time for decomposition

↓ DEPTH · OLDER

Part 2 – Carbon measurement projects



Day 1

FOUR STEPS TO COMPLETING A CARBON PROJECT IN AN ECOSYSTEM

01

Sampling design

Where, how many, and when to sample.

02

Obtaining cores

Drive, extract and section the sediment core.

03

Sample analysis

Bulk density and organic carbon in the lab.

04

Calculating carbon

From subsection to core to whole study area.

PART 2 - SECTION 01

SAMPLING DESIGN

Sampling design determines where, how many, and when to sample to produce a reliable estimate.

What is a sampling design?

Connects field work to intended outcomes

How?

- Provides a framework for choosing what and where to sample to estimate the carbon stored in an area
- Sampling designs allow for the strategic measurement of plots and sites within a larger study area
- Study sites are identified to capture variation of the area
- Plots are laid out within sites to obtain a representative sample of the study site

STUDY AREA

THE ENTIRE AREA THAT YOU
WISH TO INVESTIGATE



STUDY SITES

IDENTIFIED TO CAPTURE THE
VARIATION OF THE AREA



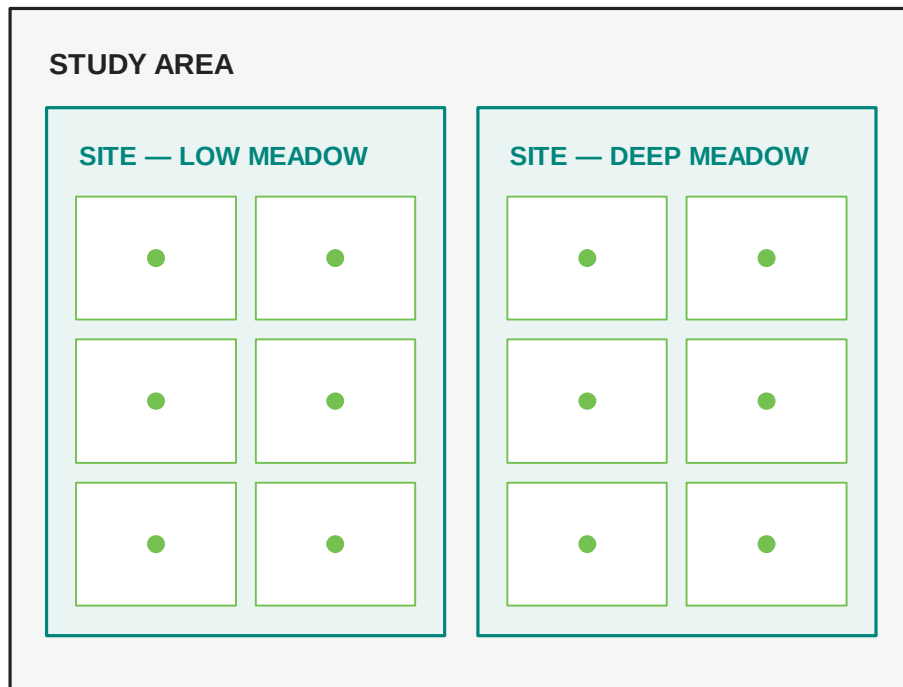
PLOTS

WHERE SAMPLING
TAKES PLACE



SAMPLING DESIGN

STUDY AREA, SITES & PLOTS



Study area

The entire area you want a carbon estimate for.

Sites

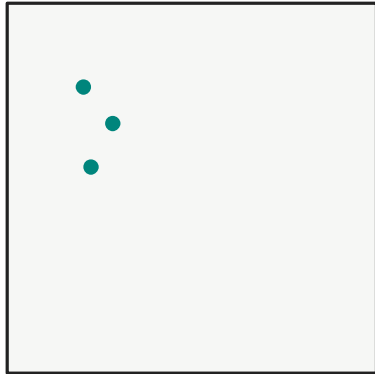
Smaller, more uniform zones that capture the area's variation — split by depth, vegetation, or management.

Plots

Where the coring happens. Combine plot measurements to estimate the whole.

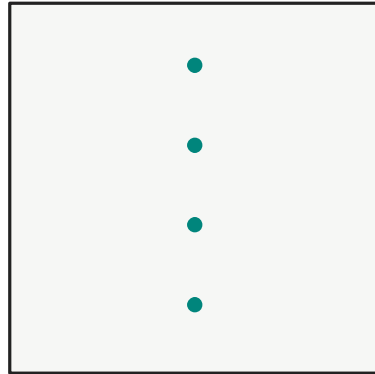
SAMPLING DESIGN

SAMPLING STRATEGY



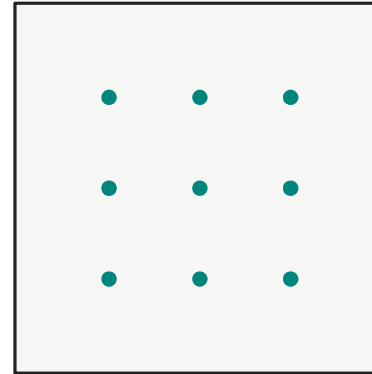
CONVENIENT

Easy access; weak coverage.



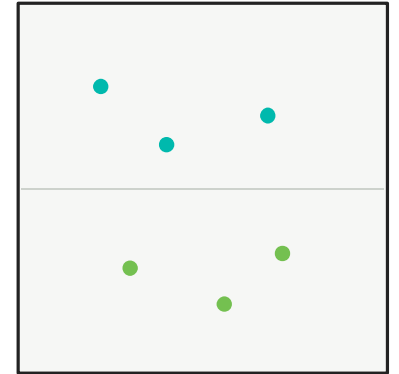
LINEAR

Transect across a gradient.



GRID

Even, systematic coverage.



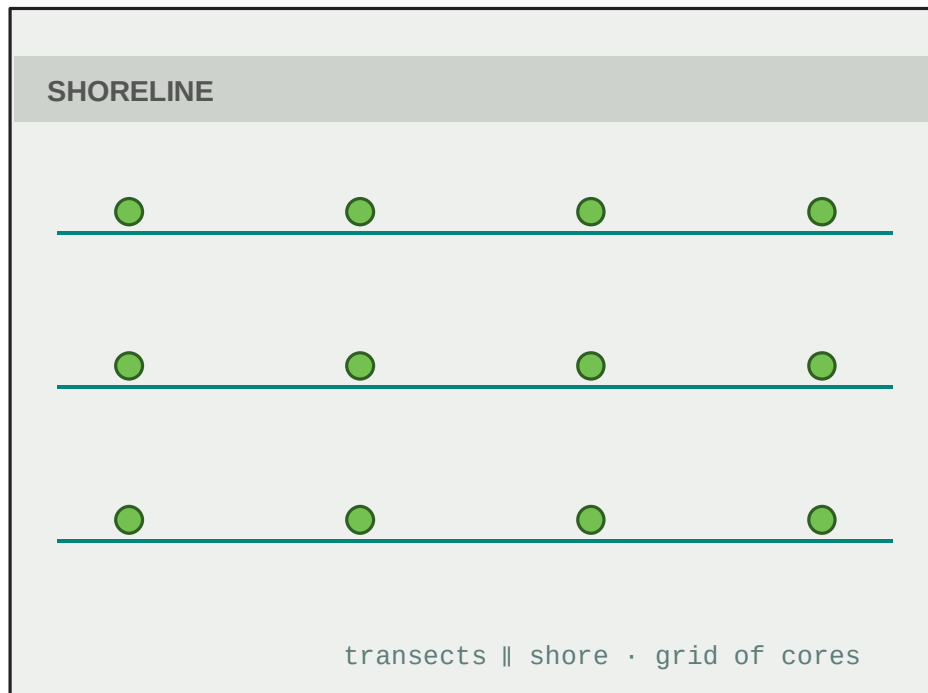
STRATIFIED-RANDOM

Random within each zone.

No single strategy is best — pick the one that captures your meadow's variation for the effort you can afford.

SAMPLING DESIGN

CONSIDERATIONS



- ↔ **Transects parallel to shore**
Each transect stays at a similar depth and sediment type.
- **Grid within each site**
A random or probability grid spreads cores representatively.
- ×2 **At least two replicates**
Two or more cores per site let you estimate variation.
- ↓ **Work at low tide**
Less standing water means cleaner, less-disturbed cores.

SAMPLING DESIGN

Considerations - Timing

When to go?

SEASON Spring to fall — avoid snow and ice.

TIDES Low tide — easier access, less water disturbance.

GROWTH If sampling plants too, aim for peak growth.

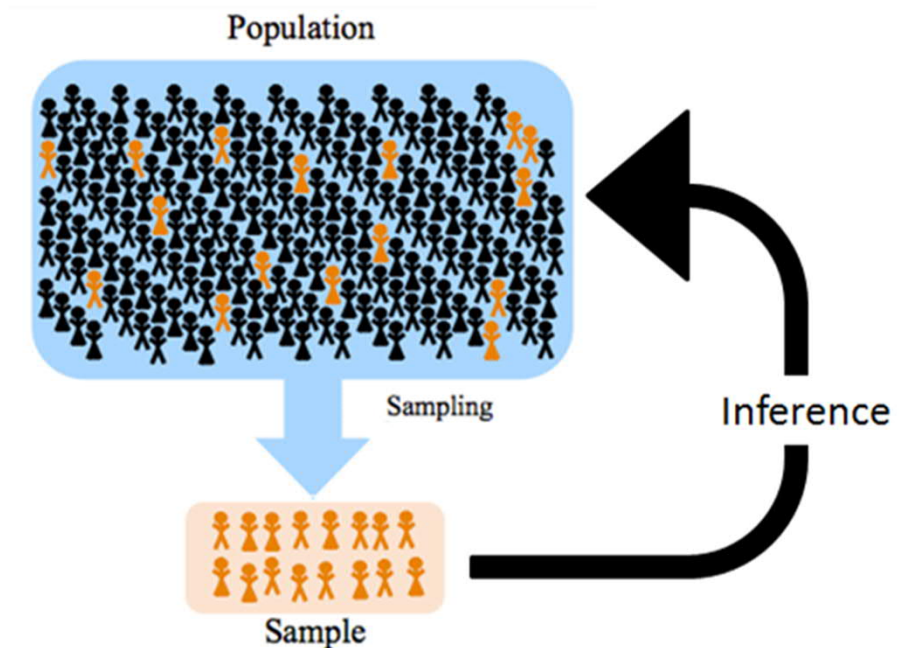
PLAN AHEAD

If applicable - confirm permits and site access before the field day — alternative sites take time to arrange.

How many samples to collect?

Determine how many samples to take

- By sampling an ecosystem, we are selecting a small portion of the total area to approximate the value of the larger area
- There is no one-size-fits-all formula, but three factors will influence your number of samples:
 1. The size of your study area
 - Greater than 10,000 ha
 2. Project goals
 - Desired accuracy
 3. Budget



How many samples to collect?

Required Inputs:

- **Size of study area (m²)**
- **Allowable error** (default = 20%)
- Desired level of **precision** (default = 10%)

Optional:

- Mean
- Standard deviation

CarbonMonitoring_SampleDesignGuide_SampleAllocationCalculator		
File Edit View Insert Format Data Tools Extensions Help		
Q Menus ↶ ↷ 🖨 🔍 100% \$ % 0.0 123 Arial 11		
K8		
	A	B
1	This spreadsheet will calculate the mininum expected sample allocation	
2	Part 1 - For total study area	
3	Things to be input my user:	Value
4	a) Size of total study area (m^2)	10000
5	b) If stratified, how many strata	1
6	c) Allowable error	0.1
7	d) Alpha value	0.1
8		
9	Preset Values (can be updated by user)	Value
10	Forest Carbon Mean Value (from wwf-canada carbon map)	4.14
11	Forest Carbon SD Value (from wwf-canada carbon map)	1.8
12	Soil Carbon Mean Value (wwf-canada carbon map)	13.2
13	Soil Carbon SD Value (wwf-canada carbon map)	10
14	Z-statistic (soils)	1.644853625
15	Z-statistic (forests)	1.644853625
16		
17	Output	
18	Forests (n) - estimated number of plots in study area	16.79192071
19		

How many samples to collect?

Required Inputs:

- **Size** of study area (m²)
- **Allowable error** (default = 20%)
- Desired level of **precision** (default = 10%)

Optional:

- Mean
- Standard deviation

Step 3: Calculate Stratified Sample Size

Confidence Level (%):

Margin of Error (%):

Plot Type:

Sediment Core (100 m²)

 Calculate Stratified Sample Size

✓ Stratified Sample Size Calculated

Total Sample Size: 12 plots

Confidence: 80% | Error: ±30%

t-value (df=10): 1.383

Margin of Error: ±3.000 kg/m²

Iterations: 5 (converged)

Plot Size: 0.01 ha (100 m²)

Proportional Allocation by Stratum

Stratum	Plots	%	σ
Eelgrass2	6	50.0%	7.50
Eelgrass-1	6	50.0%	7.50



Designing a Carbon Project

5 STEPS

01 Define study area boundaries

02 (Optional) Stratify the area (divide into different areas)

03 Select carbon pools (What to measure)

04 Determine sample allocation (How many samples)

05 Determine sample distribution (Where to take samples)

Designing a Blue Carbon Project

01 Define study boundaries

- Prior maps
- Digital maps
- Rough outlines

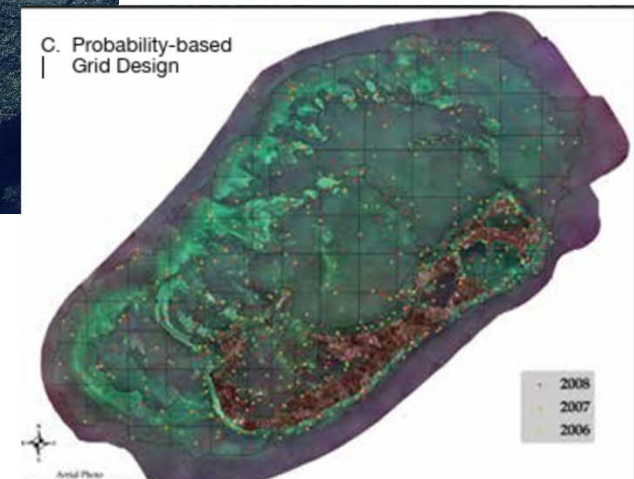


Designing a Blue Carbon Project

02 (Optional) Stratify the area (divide into different areas)

Based-on

- Ecosystem type
- Management zone
- Community interest
- Temporally



Designing a Blue Carbon Project

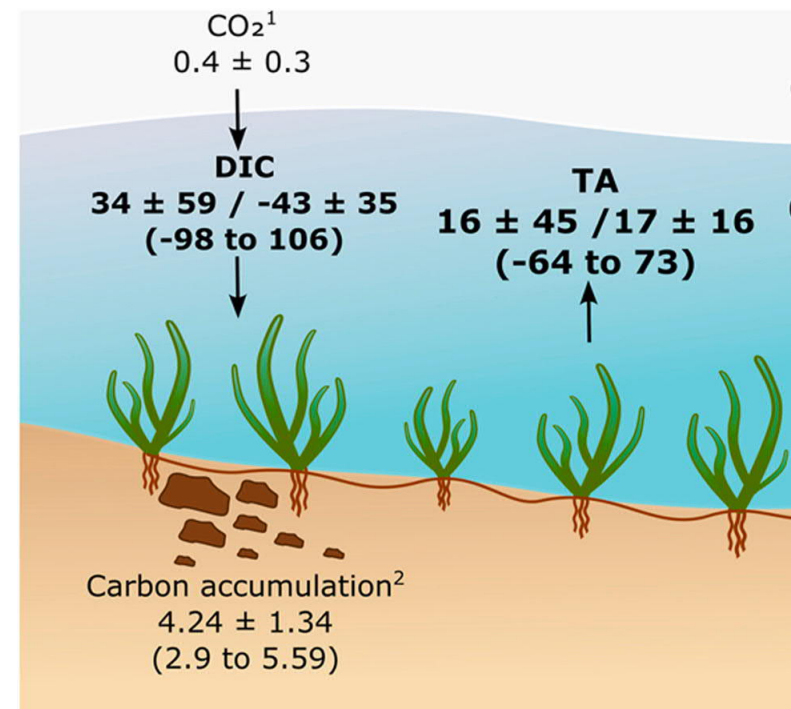
03 Select carbon pools (What to measure)

Biomass

- Above-ground (Shoots)
- Below-ground (Roots)

Sediment carbon (Coring)

Dissolved in Water (DIC, DOC)



Designing a Blue Carbon Project

04 Determine sample allocation (How many samples)

Area = 20 ha

Accuracy = 90%

Margin of Error = 20%

Variation = 0.75

Step 3: Calculate Stratified Sample Size

Confidence Level (%):

Margin of Error (%):

Plot Type:

Sediment Core (100 m²)

 Calculate Stratified Sample Size

✓ Stratified Sample Size Calculated

Total Sample Size: 12 plots

Confidence: 80% | Error: ±30%

t-value (df=10): 1.383

Margin of Error: ±3.000 kg/m²

Iterations: 5 (converged)

Plot Size: 0.01 ha (100 m²)

Proportional Allocation by Stratum

Stratum	Plots	%	σ
Eelgrass2	6	50.0%	7.50
Eelgrass-1	6	50.0%	7.50



Designing a Blue Carbon Project

05 Determine sample distribution (Where to take samples)

a) Convenient

b) Random

c + d) Systematic

(a)



(b)



(c)

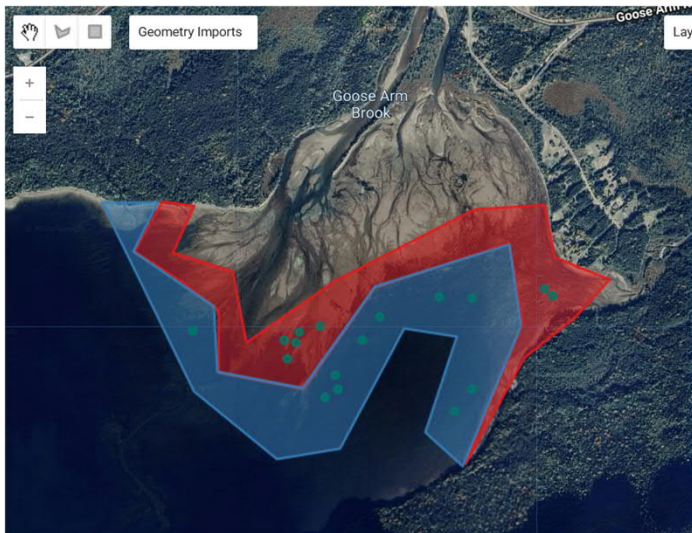


(d)

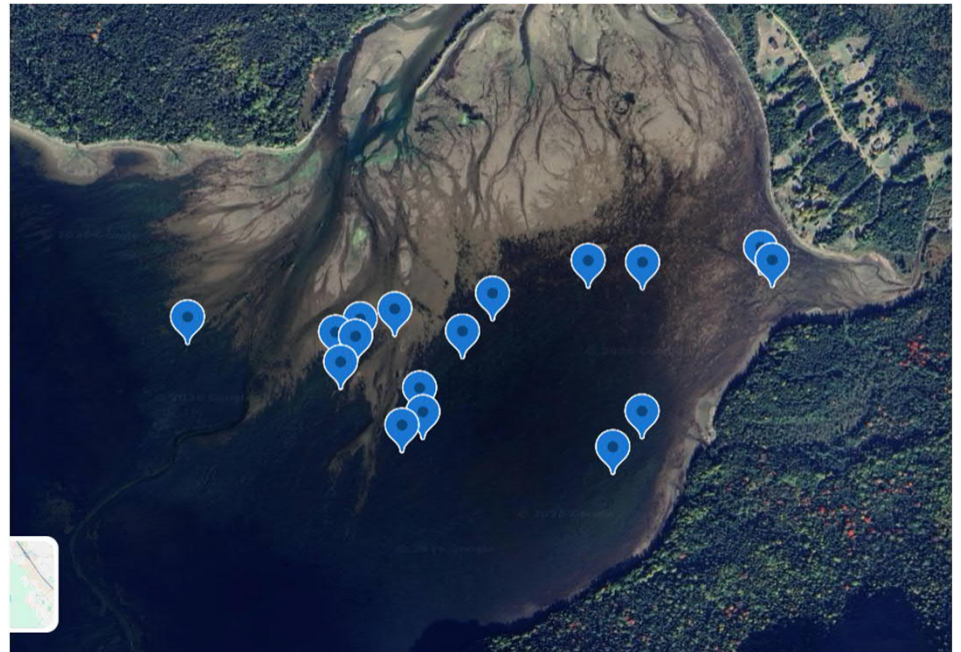


Designing a Blue Carbon Project

05 Determine sample distribution (Where to take samples)



Export to KML



Part 2 - SECTION 02

OBTAINING CORES

Driving, extracting and sectioning the open-barrel core.

Measuring sediment carbon

Collecting samples – considerations

- Sediments are often very wet
- Need to consider tides
- Subtidal seagrass areas may need a dive team
- Seagrass sediments may have shell hash (inorganic C)

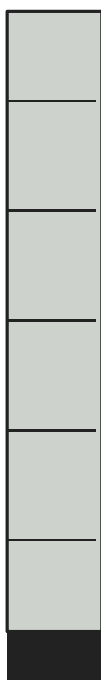
Methods

- PVC push corer is often used for soft seagrass meadow sediment
- 3" or 4" PVC pipe, sharpened at end, pushed into sediment



OBTAINING CORES

THE OPEN-BARREL CORER



Headspace (~20 cm)

Empty space above the sediment so the core isn't squeezed.

Ruler markings

Drawn on the outside so you can read insertion depth.

Open barrel

3–4" tube, as long as your target depth.

Bevelled cutting end

Sharpened edge slices cleanly into the sediment.

FIELD KIT

Open-barrel tube + caps	Tape measure (1 m)
Sledgehammer	GPS unit
Plywood / lumber	Datasheet + pencil
Work gloves	Ziploc bags / jars
Shovel	Cooler + ice

Extruding device, collar and serrated knife come in for sectioning.

OBTAINING CORES

LOCATION & DEPTH OF REFUSAL

(Optional) Probe depth (cm) across the plot — core near the average

52	61	58	49	63
55	57	60	54	62
59	53	58	56	51

■ Average $\approx$ 57 cm $\rightarrow$ core here

Find a representative spot

Probe sediment depth on a 1 m grid across a 10 $\times$ 10 m area, average the readings, and core where the depth is closest to that average.

DEPTH OF REFUSAL

The point where the corer can't be pushed any further — the base of the carbon-rich organic layer.

Core to refusal whenever you can — it's the only way to capture the full carbon stock.

OBTAINING CORES

INSERTING THE CORER

STEP

1

Place & plumb

Open barrel bevelled-end down, kept as vertical as possible.

2

Seat by hand

Push down until the barrel sets firmly and stands on its own.

3

Drive it in

Lumber across the top; hammer in steady increments, staying straight.

4

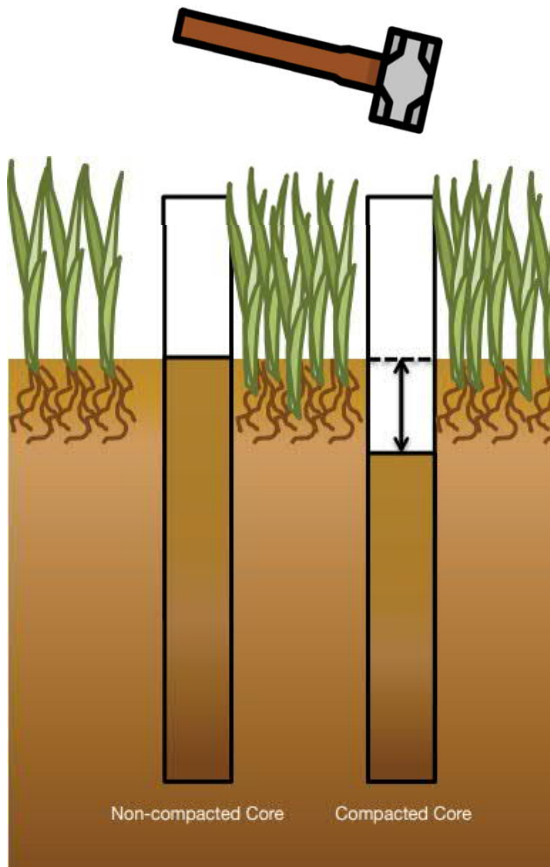
Track the depth

Read the ruler marks; drive to refusal or your target depth.



OBTAINING CORES · KEY MEASUREMENT

MEASURING COMPACTION



As you hammer, sediment compresses inside the barrel. Measure it both ways:

In the field

Compaction = (top-of-barrel → ground surface) – (top-of-barrel → sediment inside).

$\text{ratio} = \text{recovery} \div \text{drive}$

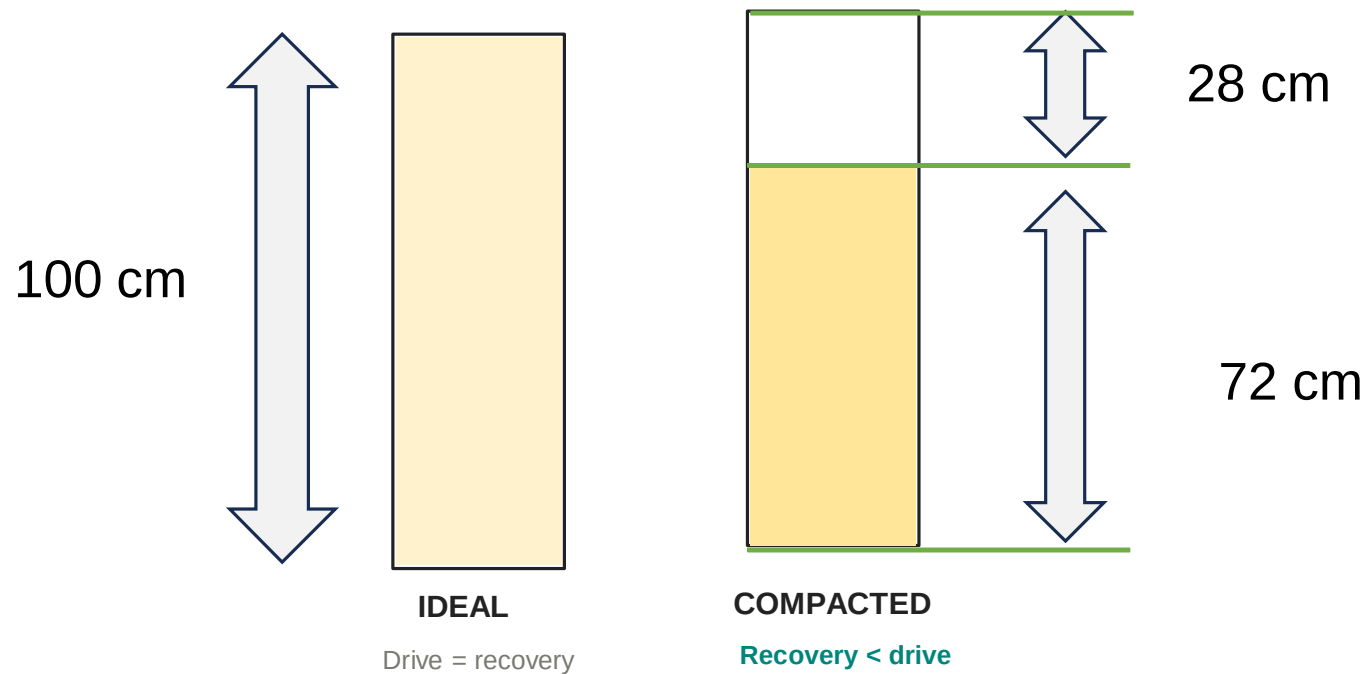
$\text{corrected interval} = \text{measured} \div \text{ratio}$

0.72 Recover 72 cm from a 100 cm drive → ratio 0.72. Every slice's depth is stretched back out by this factor.

Record drive & recovery for every core — without them, depths can't be corrected.

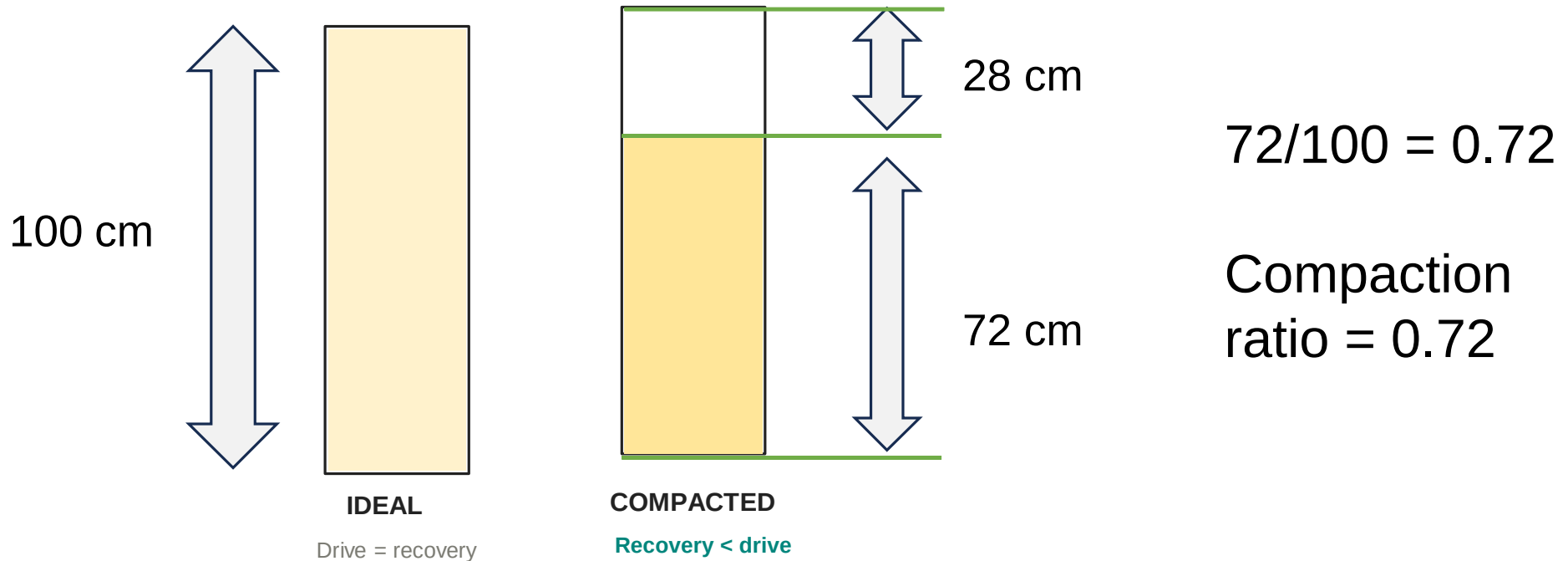
OBTAINING CORES · KEY MEASUREMENT

MEASURING COMPACTION



OBTAINING CORES · KEY MEASUREMENT

MEASURING COMPACTION



OBTAINING CORES

EXTRACTING THE CORE

1 Cap the top first

A top cap holds the sediment in place by suction before you lift.

2 Break the suction

Dig alongside the barrel with a shovel, or rock it gently back and forth.

3 Cap the bottom

Once the base is exposed, cap it so no sediment is lost on the way up.

4 Keep it upright

Carry the core vertical so layers don't mix before sectioning.

SEAGRASS NOTE

In waterlogged or rocky beds you may lose the bottom — if so, pick a new spot and try again.

OBTAINING CORES

EXTRUSION & SECTIONING

0-2
2-4
4-6
6-8
8-10

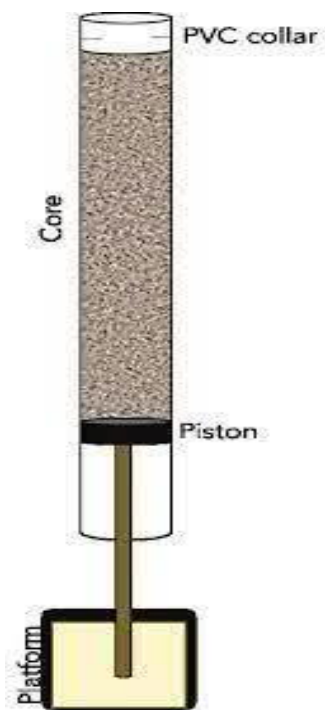


UC-02-B · 0-2 cm

UC-02-B · 2-4 cm

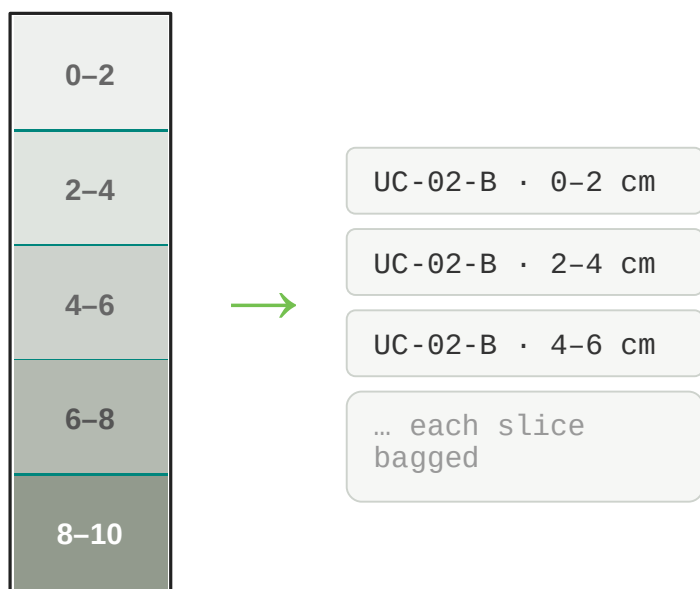
UC-02-B · 4-6 cm

... each slice
bagged



OBTAINING CORES

EXTRUSION & SECTIONING



Extrude

Slide the core onto the piston of the extruding device and push it up from the base.

Slice in 2 cm sections

Set the collar on top, push up, and cut with a serrated knife — it parts roots cleanly.

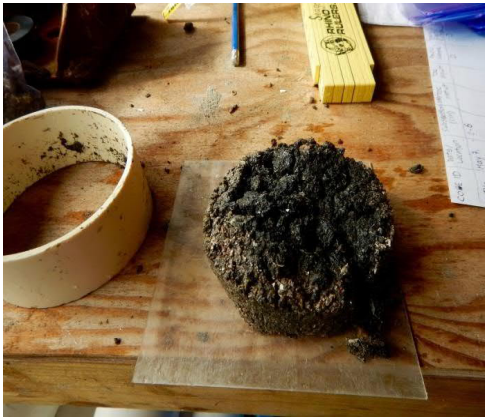
Bag & label

Each slice into its own pre-labelled bag with a unique Core ID and depth.

Core ID = site · plot · location · depth → **UC-02-B: 0-2 cm**

OBTAINING CORES

PACKAGING SAMPLES



FIELD

Cooler

Bagged slices go straight into a cooler with ice.

OBTAINING CORES

PACKAGING SAMPLES

FIELD

Cooler

Bagged slices go straight into a cooler with ice.



STORAGE

Freezer -20 °C

Transfer to a freezer as soon as possible for long-term storage.



PRE-LAB

Thaw 5–6 °C

Thaw slowly in a fridge for 2–5 days before processing.

Why cold? Warm sediment keeps respiring — microbes break down organic matter within hours and shift your carbon reading.

PART2 - SECTION 03

SAMPLE ANALYSIS

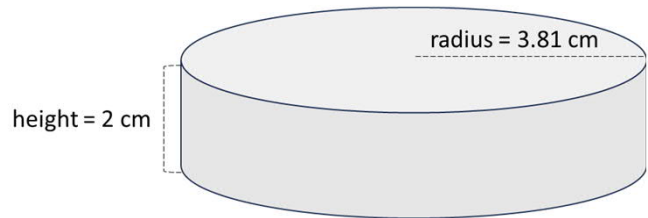
Bulk density and organic carbon, slice by slice.

SAMPLE ANALYSIS

DRY BULK DENSITY

Bulk density combines weight and volume

$$\text{bulk density} = \frac{\text{dry mass (g)}}{\text{volume (cm}^3\text{)}}$$



SLICE VOLUME

$$V = \pi \times r^2 \times \text{thickness}$$

Example: 3" tube ($r = 3.81$ cm), 2 cm slice:

$$V = \pi \times 3.81^2 \times 2 \approx 91.2 \text{ cm}^3$$

*A 20% error in bulk density is a 20% error in your carbon stock, always measure, rather than estimate, bulk density.

SAMPLE ANALYSIS

ORGANIC CARBON

LOI₅₅₀ — LOSS ON IGNITION

Burn ~1 g of dried, ground sediment at 550 °C for 4 hours. The weight lost is organic matter — a low-cost proxy for organic carbon.

ELEMENTAL ANALYSIS — %OC

A CHN analyser measures organic carbon directly. More accurate — preferred where carbonate (shell hash) is present.

Mind the conversion

Turning LOI into %carbon uses a conversion factor — but it's not universal. Eelgrass sediments can differ from the textbook value.

BEST PRACTICE

Run paired LOI + elemental analysis on a subset to calibrate the factor for your own meadow before applying it everywhere.

Always prep dry, homogenised sediment — grind until no distinct bits remain.

Measuring sediment carbon

Collaboration with partners

- Samples from the same area can be used for different analyses, e.g. carbon, heavy metals, pesticides, etc.
- Sharing equipment and working together to collect samples can reduce labour and lab costs
- University labs may be able to assist with different analyses



PART 2 - SECTION 04

CALCULATING CARBON

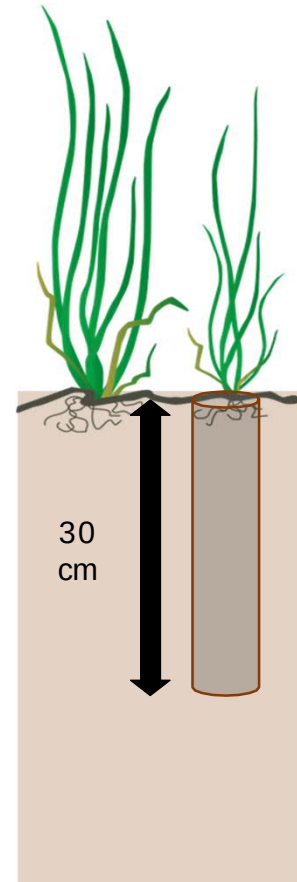
From a single slice up to the whole study
area.

Recall: Carbon Stocks

- the amount of carbon stored in a reservoir or pool (e.g. sediment) at a given time
- Combining dry bulk density with % carbon will give you the

Sediment carbon stocks
 gC / cm^3

Over a given depth (e.g. the top 30 cm)



CALCULATING CARBON

SUBSECTION → CORE

STEP 1 · CARBON DENSITY OF A SLICE

$$\text{density (g/cm}^3\text{)} = \text{bulk density} \times (\%C \div 100)$$

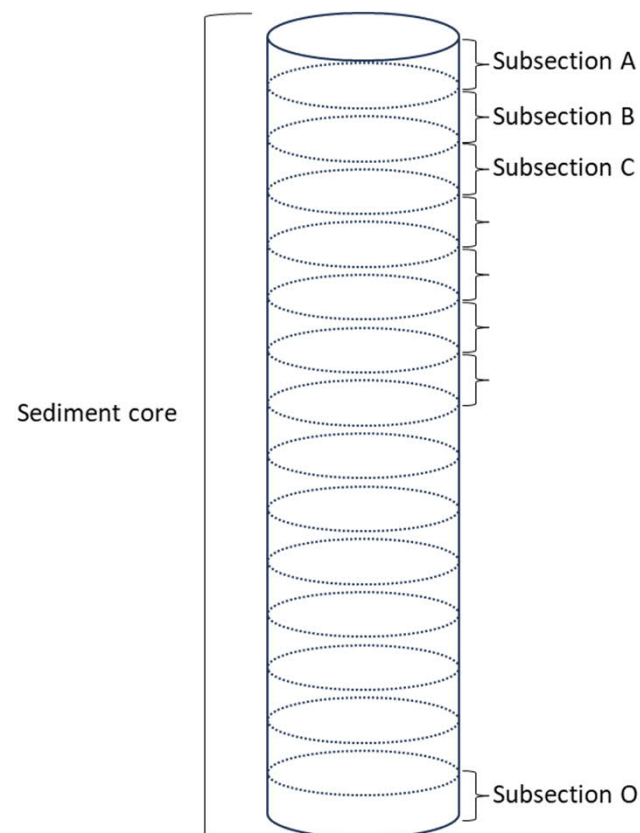
STEP 2 · CARBON IN THAT SLICE

$$\text{slice stock (g/cm}^2\text{)} = \text{density} \times \text{slice thickness (cm)}$$

STEP 3 · SUM THE WHOLE CORE

$$\text{core stock} = \sum \text{ slices} \rightarrow \times 10 = \text{kg/m}^2$$

Multiplying g/cm² by 10 converts to kg/m². The slices must cover 100% of the core.



QUALITY ASSURANCE

QUALITY ASSURANCE — MOST COMMON DATA FAILURES

Not recording compaction

Depths can't be corrected — stock is wrong.

Crushing slices when cutting

Ruins bulk density — that slice is lost.

Mixing up core / sample IDs

A whole core can become unusable.

Warm transport, no ice

Microbes inflate the carbon reading.

Coring on disturbed ground

Wrack or scars give unrepresentative carbon.

Do: measure, don't borrow

Use your own bulk density for every core.

BEFORE YOU GO

PRE-DEPARTURE CHECKLIST

☐ Corer, caps, hammer, lumber, shovel

☐ GPS

☐ Cooler (Ideally)

☐ Freezer space & lab drop-off arranged

☐ Datasheets on waterproof paper

☐ Tide times checked — coring window planned

☐ Sample bags pre-labelled with Core IDs

☐ Permits, access & safety plan filed

IN CLOSING

KEY TAKEAWAYS

- 01** Core to the **depth of refusal** — it's the only way to capture the full stock.
- 02** Record **compaction** on every core, or the depths can't be corrected.
- 03** Maintain consistent sample IDs throughout field and lab
- 04** Measure volume and bulk density accurately — never estimate
- 05** Your results are only as strong as your field methods.

PART 2 - SECTION 06

Data Interpretation

From scattered cores to a continuous
carbon map.

CALCULATING CARBON

CORE → STUDY AREA

CORE

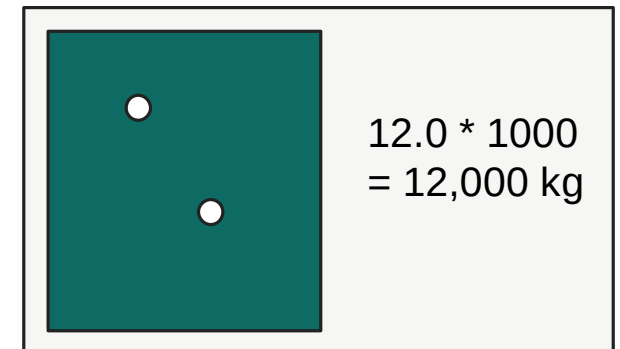
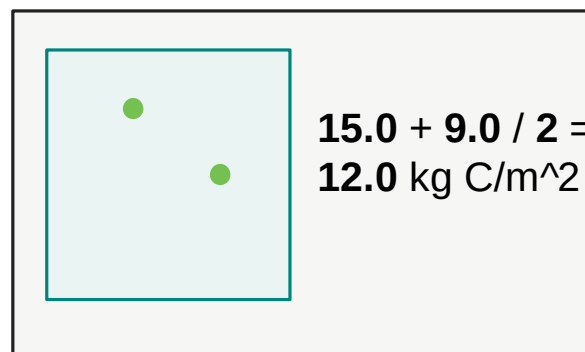
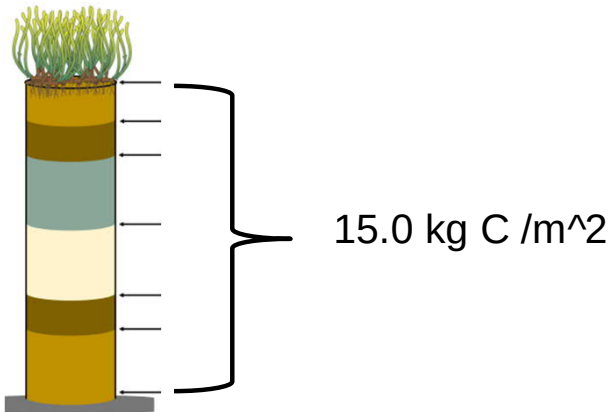
Stock of each core, in kg/m²

SITE

Average the cores → × site area
for the site total

STUDY AREA

Combine sites → × total area for
the whole stock



Option 1 – Simple area-based extrapolation

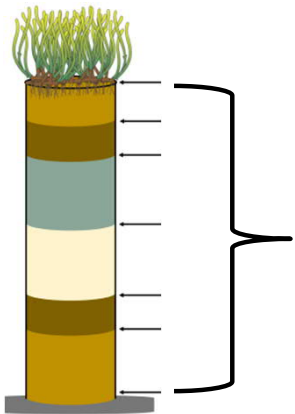
INPUTS

Core stocks + Area

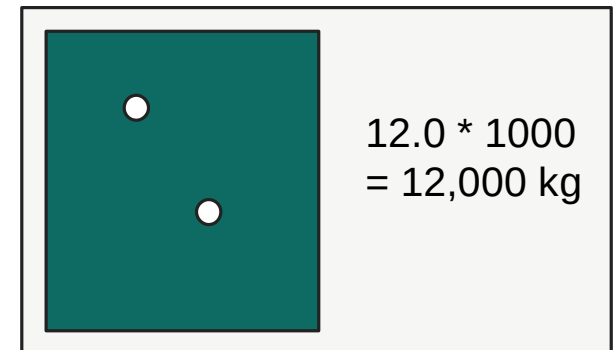
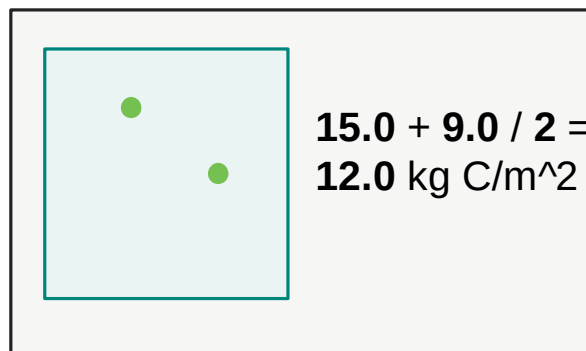


CARBON MAP

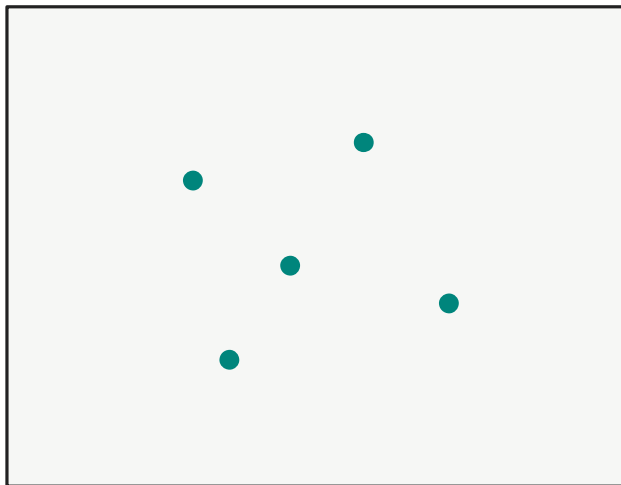
Predicted values



15.0 kg C /m²



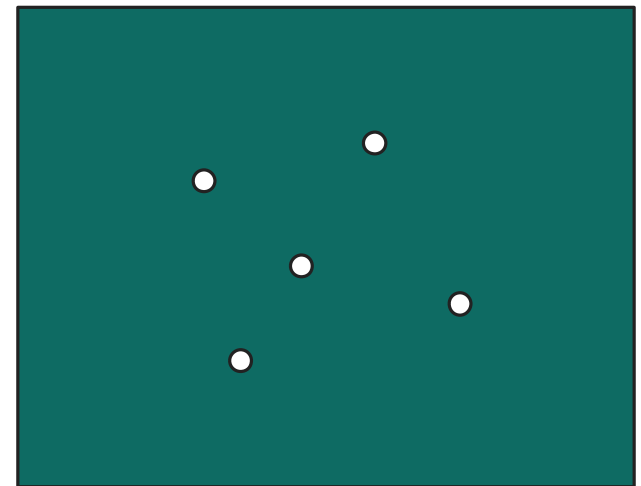
Option 1 – Simple area-based extrapolation



A FEW CORES

Carbon measured at points.

Average all cores
in an area



A CONTINUOUS MAP

1 value for each area averaging
each core in the area.

BEYOND THE CORE

Option 2 – Modelling within areas of interest

INPUTS

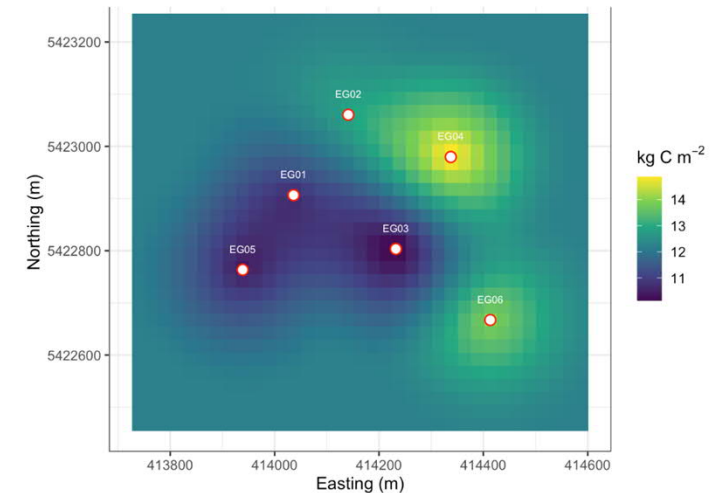
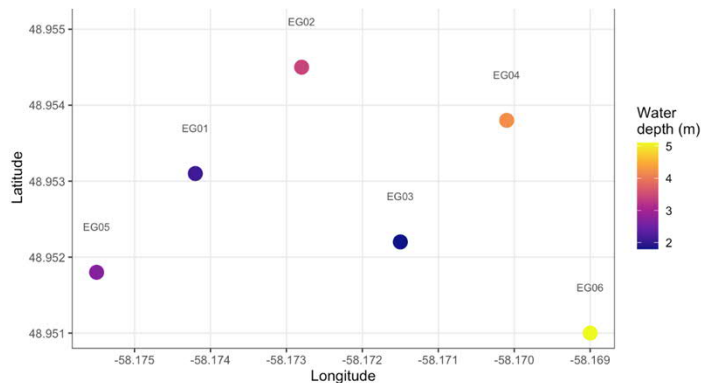
Core stocks + Area

Extrapolation

Estimate Carbon values at every point

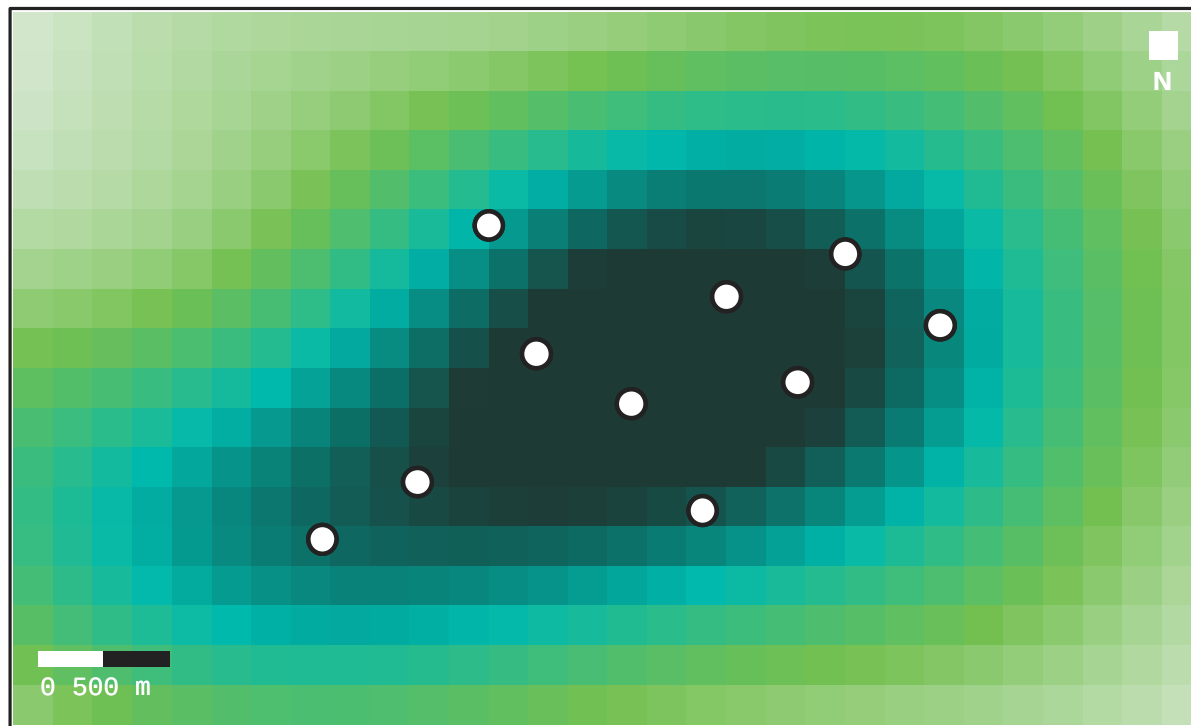
CARBON MAP

Predicted values



BEYOND THE CORE · EXAMPLE OUTPUT

THEMATIC CARBON MAP



Low  high Predicted carbon stock (kg C/m²)

○ Measured core locations

The model paints a predicted value into every grid cell, anchored by the real cores.

EXPORT

Save as a **GeoTIFF** to open in **Google Earth Engine** or **ArcGIS** — for mapping, zonal totals and reporting.

Pair the map with an uncertainty layer so each estimate carries its confidence.

SECTION 06

DATA ANALYSIS WORKFLOW

R-based workflow for processing sediment core data from field collection to carbon stock estimates.

DATA WORKFLOW

Downloading the files and setting up the analysis

Link to GitHub with code and instruction for data interpretation:

https://github.com/CathalD/BlueCarbon_Eelgrass_Workshop.git